Hamiltonian Thresholds for Planar Recolouring Graphs

October 2026

Abstract

Let $\mathcal{R}_k(G)$ be the graph of all proper colourings of a finite graph G with a labelled palette $[k]$, with one-vertex recolourings as edges. We collect sharp Hamiltonian and Hamilton-connected thresholds for planar graph classes. A d-degenerate graph has a Hamilton-connected and vertex-pancyclic recolouring graph for every $k \geq d+3$. The resulting Hamilton-connected thresholds are 7 for all planar graphs, 5 for triangle-free planar graphs (and hence for bipartite planar graphs), 4 for planar graphs of girth at least six, for nonempty forests, and for the single graph C_5. We give the exact Hamiltonian thresholds 4 for planar girth-five graphs with at most seven distinct 5-cycles, for maximum degree at most three. For these girth-five classes, four-colour Hamilton-connectedness remains open. The general planar girth-five four-colour problem remains open: the current bounds are $4 \leq h \leq 5$ and $4 \leq \mathrm{hc} \leq 5$.

1 Introduction

All graphs in this paper are finite and simple; they may be disconnected. A forest is assigned infinite girth. Colours are labelled, and a proper colouring need not use every colour. All graph classes in the threshold definitions below consist of nonempty graphs. If G is a graph, then $\mathcal{R}_k(G)$ has as vertices the proper maps $\sigma : V(G) \to [k]$, and two colourings are adjacent when they differ on exactly one vertex of G.

A graph is Hamiltonian if it has a spanning cycle, Hamilton-connected if every two distinct vertices are joined by a spanning path, and vertex-pancyclic if every vertex lies on a cycle of every length between three and the order of the graph. For a class $\mathcal{C}$ of nonempty graphs define

$$h(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamiltonian for all } G \in \mathcal{C},\ k \geq q\},$$

$$\mathrm{hc}(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamilton-connected for all } G \in \mathcal{C},\ k \geq q\},$$

whenever the displayed set is nonempty. The word threshold always has this uniform meaning; no monotonicity in k is assumed.

The principal conclusions are summarized in Table A.1 in Appendix A. A range such as $4 \leq \mathrm{hc} \leq 5$ records a proved four-colour lower bound and a proved five-colour upper bound, with the four-colour Hamilton-connected statement unresolved.

The four-colour recolouring graph of every planar graph of girth at least five is connected by Bartier et al. [5]. Earlier recolouring and Gray-code work gives useful connectivity context [1, 2], but the Hamilton-connected strengthening used here is separate. Connectivity, a Hamilton cycle in a restricted fibre, and a finite state computation do not imply a Hamilton cycle of the complete recolouring graph.

2 The degenerate theorem

We begin with the general result that supplies all large-palette statements.

Lemma 2.1 (Cyclic list selection)**.** *Let $m \geq 3$ and let $L_1, \ldots, L_m$ be finite sets of size at least two. Unless m is odd and all L_i are the same two-element set, there are $x_i \in L_i$ with $x_i \neq x_{i+1}$ cyclically.*

Proof. If some L_i has at least three elements, choose successively along the path obtained by deleting that position, then choose the remaining element to avoid both end values. If all sets are binary and two consecutive sets differ, start with a value in their symmetric difference and continue greedily. If all are the same binary set, alternation is possible exactly when m is even. □ □

Lemma 2.2 (One-vertex lift)**.** *Let $H = G - v \neq \varnothing$ and suppose $k \geq d_G(v) + 3$. If $\mathcal{R}_k(H)$ has a Hamilton cycle, then $\mathcal{R}_k(G)$ has a Hamilton cycle.*

Proof. Write a Hamilton cycle of $\mathcal{R}_k(H)$ as $f_1, \ldots, f_m, f_1$ and put $L_i = [k] \setminus f_i(N_G(v))$. Then $|L_i| \geq 3$ and $|L_i \cap L_{i+1}| \geq 2$. Lemma 2.1 gives cyclic choices $x_i \in L_i \cap L_{i+1}$ with consecutive values distinct. The extensions above f_i form a clique on L_i; traverse that clique by a Hamilton path from x_{i-1} to x_i, and use the edge with v fixed at x_i to cross to the next fibre. The exceptional all-identical binary case is impossible because every proper colouring of H occurs on the base cycle and a global colour permutation changes a neighbour of v to either binary colour. The fibre paths therefore concatenate to a spanning cycle. □ □

Lemma 2.3 (Bridge colours)**.** *Let $t \geq 1$, and let $S_0, \ldots, S_{t-1}$ be finite sets of colours such that $|S_i| \geq 2$ for every i. Suppose that $|S_j| \geq 3$ for some $j \in \{0, \ldots, t-1\}$. For any colours a and b, there exist colours $x_i \in S_i$ satisfying*

$$x_0 \neq a, \qquad x_i \neq x_{i-1} \quad (1 \leq i \leq t-1), \qquad x_{t-1} \neq b.$$

Proof. Set $x_{-1} = a$ and $x_t = b$. Choose $x_0, \ldots, x_{j-1}$ in this order, each time avoiding the preceding colour. Then choose $x_{t-1}, \ldots, x_{j+1}$ in this order, each time avoiding the following colour. Each of these choices excludes at most one colour from a list of size at least two. Finally choose

$$x_j \in S_j \setminus \{x_{j-1}, x_{j+1}\}.$$

At most two colours are excluded, so this choice is possible. The boundary assignments $x_{-1} = a$ and $x_t = b$ also cover $j = 0$, $j = t-1$, and $t = 1$. □

Theorem 2.4 (Degenerate Hamilton-connectedness)**.** *Let $d \geq 0$ be an integer, let G be a nonempty d-degenerate graph, and let $k \geq d + 3$ be an integer. Then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. We proceed by induction on $n = |V(G)|$. If $n = 1$, then $\mathcal{R}_k(G) \cong K_k$, which is Hamilton-connected. If $n = 2$, the edgeless graph gives $K_k \square K_k$, which is Hamilton-connected by Lemma 3.1. For $G = K_2$, we have $d \geq 1$ and hence $k \geq 4$, so the arrangement-graph result below applies. We may therefore assume that $n \geq 3$.

We next handle complete graphs $G = K_n$. The arrangement graph $A_{k,n}$ has as vertices the ordered n-tuples of distinct elements of $\{1, \ldots, k\}$, with two tuples adjacent if they differ in exactly one coordinate. Hence

$$\mathcal{R}_k(K_n) \cong A_{k,n}.$$

Since $n - 1 \leq d$, we have $k \geq d + 3 \geq n + 2$. By the Hamiltonian connectivity result of Hsu, Li, Tan and Hsu [4, Theorem 1], $A_{k,n}$ is Hamilton-connected.

We may therefore assume that G is not complete. Choose a vertex v_1 of minimum degree in G, and put

$$H := G - v_1.$$

The d-degeneracy of G gives $\deg_G(v_1) \leq d$. Since G is not complete, its minimum degree is at most $n-2$, so there exists

$$w \in V(H) \setminus N_G(v_1).$$

In particular, H is nonempty. It is again d-degenerate, so by the induction hypothesis $\mathcal{R}_k(H)$ is Hamilton-connected.

Let f be a k-colouring of H. We define

$$L(f) := \{1, \ldots, k\} \setminus f(N_G(v_1)).$$

Thus $L(f)$ is precisely the set of colours that may be assigned to v_1 while extending f to a proper k-colouring of G. Since v_1 has at most d neighbors in G, we have

$$|L(f)| \geq k - d \geq 3.$$

For each $f \in V(\mathcal{R}_k(H))$, define the clique

$$F_f := \{f(c) : c \in L(f)\}.$$

Here $f(c)$ denotes the k-colouring of G that restricts to f on H and assigns the colour c to v_1. Thus F_f is a complete graph on $|L(f)|$ vertices.

Observation. Let f and g be adjacent vertices of $\mathcal{R}_k(H)$. Then f and g differ on exactly one vertex $u \in V(H)$. If $u \notin N_G(v_1)$, then

$$L(f) = L(g).$$

If $u \in N_G(v_1)$, then the forbidden colour set for v_1 loses at most one colour and gains at most one colour, and hence

$$|L(f) \cap L(g)| \geq |L(f)| - 1.$$

In particular,

$$|L(f) \cap L(g)| \geq 2.$$

The edges between the cliques F_f and F_g are precisely the edges

$$f(c)g(c) \qquad \text{with } c \in L(f) \cap L(g).$$

Lifting claim. Let

$$P : f_0, f_1, \ldots, f_t$$

be a path in $\mathcal{R}_k(H)$ with $t \geq 1$, and put

$$S_i := L(f_i) \cap L(f_{i+1}) \qquad (0 \leq i \leq t-1).$$

Suppose that $|S_j| \geq 3$ for at least one j. Given colours

$$a \in L(f_0), \qquad b \in L(f_t),$$

there is a Hamilton path in

$$\mathcal{R}_k(G)\Big[\bigcup_{i=0}^{t} F_{f_i}\Big]$$

from $f_0(a)$ to $f_t(b)$.

Indeed, by the observation, $|S_i| \geq 2$ for every i. Lemma 2.3 therefore provides bridge colours $x_i \in S_i$ such that

$$x_0 \neq a, \qquad x_i \neq x_{i-1} \quad (1 \leq i \leq t-1), \qquad x_{t-1} \neq b.$$

Since each F_{f_i} is complete, choose Hamilton paths

$$\begin{aligned} P_0 &: \ f_0(a) \rightsquigarrow f_0(x_0) \quad \text{in } F_{f_0}, \\ P_i &: \ f_i(x_{i-1}) \rightsquigarrow f_i(x_i) \quad \text{in } F_{f_i} \qquad (1 \leq i \leq t-1), \\ P_t &: \ f_t(x_{t-1}) \rightsquigarrow f_t(b) \quad \text{in } F_{f_t}. \end{aligned}$$

For each $i \in \{0, \ldots, t-1\}$, the vertices $f_i(x_i)$ and $f_{i+1}(x_i)$ are adjacent in $\mathcal{R}_k(G)$. Concatenating the paths $P_0, P_1, \ldots, P_t$ along these bridge edges gives the required Hamilton path.

We now verify the additional hypothesis for the paths used below. Since H is k-colourable, permuting the colour names in a proper colouring shows that w takes all k colours among the vertices of $\mathcal{R}_k(H)$. In particular, $|V(\mathcal{R}_k(H))| \geq k \geq 3$. After deleting at most one colouring, at least $k-1 \geq 2$ different colours still occur at w. Thus every path containing all but at most one vertex of $\mathcal{R}_k(H)$ has a step $f_j f_{j+1}$ that changes the colour of w. Since $w \notin N_G(v_1)$, this step satisfies

$$L(f_j) = L(f_{j+1}), \qquad |S_j| = |L(f_j)| \geq 3.$$

Consequently, the lifting claim applies to every such path.

Let

$$x = \alpha(a) \qquad \text{and} \qquad y = \beta(b)$$

be two distinct vertices of $\mathcal{R}_k(G)$, where $\alpha, \beta \in V(\mathcal{R}_k(H))$, $a \in L(\alpha)$, and $b \in L(\beta)$. We will show that there is a Hamilton path from x to y in $\mathcal{R}_k(G)$.

Case 1: $\alpha \neq \beta$. Since $\mathcal{R}_k(H)$ is Hamilton-connected, there is a Hamilton path

$$\alpha = f_0, f_1, \ldots, f_t = \beta$$

in $\mathcal{R}_k(H)$. Since it contains every vertex of $\mathcal{R}_k(H)$, it satisfies the additional hypothesis of the lifting claim.

Applying the lifting claim to the path

$$f_0, f_1, \ldots, f_t$$

with the endpoint colours $a \in L(f_0)$ and $b \in L(f_t)$, we obtain a Hamilton path in

$$\mathcal{R}_k(G)\Big[\bigcup_{i=0}^{t} F_{f_i}\Big]$$

from $f_0(a) = x$ to $f_t(b) = y$. Since $f_0, \ldots, f_t$ exhaust all vertices of $\mathcal{R}_k(H)$, the cliques

$$F_{f_0}, F_{f_1}, \ldots, F_{f_t}$$

partition $V(\mathcal{R}_k(G))$. Therefore this path is a Hamilton path in $\mathcal{R}_k(G)$ from x to y.

Case 2: $\alpha = \beta$. Since $x \neq y$, we have $a \neq b$. Write

$$f^* := \alpha, \qquad c_1 := a, \qquad c_2 := b.$$

Suppose, for a contradiction, that there is no Hamilton path in $\mathcal{R}_k(G)$ from

$$x = f^*(c_1) \qquad \text{to} \qquad y = f^*(c_2).$$

We first show that F_{f^*} must have size exactly three. We then prove that one of its three vertices, namely $f^*(c_3)$, has no neighbor outside F_{f^*}. Recolouring a vertex of degree at most d in H will give a contradiction.

Claim 1. We have $|F_{f^*}| = 3$.

Suppose instead that $|F_{f^*}| \geq 4$. Since $\mathcal{R}_k(H)$ is Hamilton-connected and has at least three vertices, every edge of $\mathcal{R}_k(H)$ lies on a Hamilton cycle: indeed, if fg is an edge, then a Hamilton path from f to g together with the edge fg forms a Hamilton cycle. Choose a Hamilton cycle C in $\mathcal{R}_k(H)$ containing f^*, and let f_i and f_j be the two neighbors of f^* on C. Then $C - f^*$ is a Hamilton path from f_i to f_j in $\mathcal{R}_k(H) - f^*$.

By the observation,

$$|L(f^*) \cap L(f_i)| \geq |L(f^*)| - 1 = |F_{f^*}| - 1 \geq 3,$$

and similarly

$$|L(f^*) \cap L(f_j)| \geq 3.$$

Choose

$$p \in L(f^*) \cap L(f_i) \setminus \{c_1, c_2\},$$

and then choose

$$q \in L(f^*) \cap L(f_j) \setminus \{c_1, p\}.$$

Then

$$p \neq q, \qquad p \notin \{c_1, c_2\}, \qquad q \neq c_1.$$

Since F_{f^*} is a complete graph on at least four vertices, we can partition its vertex set into two vertex-disjoint paths, one with ends $f^*(c_1)$ and $f^*(p)$, and the other with ends $f^*(q)$ and $f^*(c_2)$. Denote these two paths by P and P' respectively. If $q = c_2$, take P' to be the single vertex $f^*(c_2)$.

The path $C - f^*$ contains all but one vertex of $\mathcal{R}_k(H)$, so it satisfies the additional hypothesis of the lifting claim. Apply that claim with endpoint colours p and q. This yields a Hamilton path

$$Q$$

in

$$\mathcal{R}_k(G)\Big[\bigcup_{f \neq f^*} F_f\Big]$$

from $f_i(p)$ to $f_j(q)$. Using the bridge edges

$$f^*(p)f_i(p) \qquad \text{and} \qquad f_j(q)f^*(q),$$

the concatenation

$$P,\ Q,\ P'$$

is a Hamilton path in $\mathcal{R}_k(G)$ from x to y, a contradiction. Hence $|F_{f^*}| = 3$.

We may therefore write

$$L(f^*) = \{c_1, c_2, c_3\}.$$

Since $|L(f^*)| \geq k - d$, this implies $k = d + 3$.

Claim 2. The vertex $f^*(c_3)$ has no neighbor outside F_{f^*}.

Suppose not. Then there is a colouring f_i adjacent to f^* in $\mathcal{R}_k(H)$ such that

$$c_3 \in L(f_i).$$

Choose a Hamilton cycle C of $\mathcal{R}_k(H)$ containing the edge f^*f_i, and let f_j be the other neighbor of f^* on C. Then $C - f^*$ is a Hamilton path from f_i to f_j in $\mathcal{R}_k(H) - f^*$.

By the observation, $|L(f^*) \cap L(f_j)| \geq 2$. Since $L(f^*) = \{c_1, c_2, c_3\}$, choose

$$q \in L(f^*) \cap L(f_j) \setminus \{c_3\}.$$

Thus $q \in \{c_1, c_2\}$.

Again, $C - f^*$ contains all but one vertex of $\mathcal{R}_k(H)$. Applying the lifting claim with endpoint colours c_3 and q, we obtain a Hamilton path

$$Q$$

through all cliques other than F_{f^*}, from $f_i(c_3)$ to $f_j(q)$.

If $q = c_2$, then

$$f^*(c_1),\ f^*(c_3),\ Q,\ f^*(c_2)$$

using the bridge edges $f^*(c_3)f_i(c_3)$ and $f_j(c_2)f^*(c_2)$ is a Hamilton path from x to y, a contradiction. If $q = c_1$, then the same construction gives a Hamilton path from y to x, and reversing it yields a Hamilton path from x to y, again a contradiction. Hence $f^*(c_3)$ has no neighbor outside F_{f^*}.

Since H is nonempty and d-degenerate, choose $u \in V(H)$ with $\deg_H(u) \leq d$. Under f^*, at least $k - d \geq 3$ colours are absent from $N_H(u)$. We can therefore choose one of these colours, say γ, different from both $f^*(u)$ and c_3. Recolouring u with γ gives a proper colouring g of H adjacent to f^*. No neighbor of v_1 has acquired the colour c_3, so $c_3 \in L(g)$. Thus $f^*(c_3)$ is adjacent to $g(c_3)$ outside F_{f^*}, contradicting Claim 2.

In both cases there is a Hamilton path from x to y. Therefore $\mathcal{R}_k(G)$ is Hamilton-connected. □

Theorem 2.5 (Degenerate vertex-pancyclicity)**.** *Let $d \geq 0$ be an integer, let G be a nonempty d-degenerate graph, and let $k \geq d + 3$ be an integer. Then the recolouring graph $\mathcal{R}_k(G)$ is vertex-pancyclic.*

Proof. We proceed by induction on $|V(G)|$.

If $|V(G)| = 1$, then $\mathcal{R}_k(G) \cong K_k$. Since $k \geq 3$, every vertex of K_k lies on a cycle of every length from 3 to k. Thus the result holds.

Assume now that $|V(G)| \geq 2$, and suppose the result holds for all d-degenerate graphs with fewer vertices.

Choose a vertex v_1 of degree at most d, and put

$$H := G - v_1.$$

For each proper k-colouring f of H, define

$$L(f) := \{1, \ldots, k\} \setminus f(N_G(v_1)).$$

Thus $L(f)$ is the set of colours that can be assigned to v_1 while extending f to a proper k-colouring of G. Since $|N_G(v_1)| \leq d$, we have

$$|L(f)| \geq k - d \geq 3.$$

For each $f \in V(\mathcal{R}_k(H))$, define

$$F_f := \{f(c) : c \in L(f)\}.$$

Then F_f is a clique in $\mathcal{R}_k(G)$.

Let

$$X := \mathcal{R}_k(G), \qquad Y := \mathcal{R}_k(H).$$

The sets F_f, where $f \in V(Y)$, form a partition of $V(X)$.

As observed in the preceding proof, if f and g are adjacent in Y, then $|L(f) \cap L(g)| \geq 2$, and the edges between F_f and F_g are precisely $f(c)g(c)$ with $c \in L(f) \cap L(g)$.

Now fix an arbitrary vertex of X,

$$x = \alpha(a),$$

where $\alpha \in V(Y)$ and $a \in L(\alpha)$. We prove that x lies on a cycle of every possible length.

Let

$$M := |V(Y)|, \qquad N := |V(X)|.$$

Closed lifting claim. Let

$$C : f_0, f_1, \ldots, f_{s-1}, f_0$$

be a cycle in Y with $s \geq 3$ and $f_0 = \alpha$. Then, for every integer ℓ satisfying

$$2s + 1 \leq \ell \leq \sum_{i=0}^{s-1} |F_{f_i}|,$$

there is a cycle in X through x of length ℓ using only vertices from the cliques $F_{f_0}, F_{f_1}, \ldots, F_{f_{s-1}}$.

For i modulo s, put

$$S_i := L(f_i) \cap L(f_{i+1}).$$

We first show that there are bridge colours $y_i \in S_i$ such that $y_{i-1} \neq y_i$ for every i modulo s.

We use an elementary fact about lists on a cycle. When all lists have size at least two, a proper choice of colours exists unless the cycle is odd and all lists are the same two-element set. To see this, if some list has size at least three, leave it until last and greedily colour the path formed by the remaining positions. The last position has at most two forbidden colours. If all lists have size two but are not identical, choose adjacent positions whose lists differ, say S_r and S_{r-1}, and start with a colour in $S_r \setminus S_{r-1}$. Greedily colour the positions $r+1, \ldots, r-1$ in cyclic order. The last colour automatically differs from the first, since the first colour is absent from S_{r-1}. Finally, if all lists are the same two-element set and the cycle is even, alternate the two colours.

It remains to exclude the exceptional odd-cycle configuration in our setting. Suppose that s is odd and

$$S_0 = S_1 = \cdots = S_{s-1} = \{p, q\}.$$

For every i, the observation gives $2 = |S_i| \geq |L(f_i)| - 1$, so $|L(f_i)| = 3$. Consequently,

$$k - 3 = |f_i(N_G(v_1))| \leq \deg_G(v_1) \leq d \leq k - 3.$$

All inequalities are equalities. Thus $k = d + 3$, the vertex v_1 has exactly d neighbors, and these neighbors receive pairwise distinct colours under every f_i. None of these colours is p or q.

Every step $f_i f_{i+1}$ changes a vertex in $N_G(v_1)$; otherwise $L(f_i) = L(f_{i+1})$ would imply $|S_i| = 3$. In particular, $d \geq 1$. Fix an ordering $u_1, \ldots, u_d$ of $N_G(v_1)$, and let

$$A := \{1, \ldots, k\} \setminus \{p, q\}.$$

For each i, exactly one colour $r_i \in A$ is absent from $f_i(N_G(v_1))$. Hence

$$(f_i(u_1), \ldots, f_i(u_d), r_i)$$

is a permutation of the $d+1$ colours in A. At the step $f_i f_{i+1}$, the changed neighbor must receive r_i, because the other neighbor colours remain fixed and pairwise distinct. Its old colour becomes r_{i+1}. Thus this step exchanges the entries at one neighbor position and the last position, reversing the parity of the permutation. Since $f_s = f_0$, an odd number of such exchanges cannot return to the initial permutation. This contradiction excludes the exceptional configuration, and proves the existence of the colours y_i.

The integer intervals

$$[3, |F_{f_0}|],\ [2, |F_{f_1}|], \ldots, [2, |F_{f_{s-1}}|]$$

have sum equal to the full integer interval $[2s+1, \sum_i |F_{f_i}|]$. We may therefore choose integers $m_0, \ldots, m_{s-1}$ with

$$3 \le m_0 \le |F_{f_0}|, \qquad 2 \le m_i \le |F_{f_i}| \quad (1 \le i \le s-1), \qquad \sum_{i=0}^{s-1} m_i = \ell.$$

Inside F_{f_0} take a path from $f_0(y_{s-1})$ to $f_0(y_0)$ on m_0 vertices and containing x. For $1 \le i \le s-1$, take inside F_{f_i} a path from $f_i(y_{i-1})$ to $f_i(y_i)$ on m_i vertices. The inequalities $y_{i-1} \ne y_i$ make these paths possible, and the bridge edges $f_i(y_i)f_{i+1}(y_i)$, with indices modulo s, close them into the desired cycle.

Since $|L(\alpha)| \ge 3$, the clique F_α contains a triangle through x.

Next we show that x lies on cycles of lengths $4, 5, 6$. Since H is nonempty and d-degenerate, choose a vertex $u \in V(H)$ of degree at most d in H. Because $k \ge d+3$, we can recolour u to some colour different from its present colour, avoiding all colours on $N_H(u)$ and, if $u \in N_G(v_1)$, also avoiding the colour a. Thus there is a colouring β adjacent to α in Y such that

$$a \in L(\beta).$$

Since

$$|L(\alpha) \cap L(\beta)| \ge 2,$$

choose

$$p \in L(\alpha) \cap L(\beta), \qquad p \ne a.$$

Also choose

$$q \in L(\alpha) \setminus \{a, p\}$$

and

$$r \in L(\beta) \setminus \{a, p\}.$$

Then the following are cycles in X through x:

$$\alpha(a), \alpha(p), \beta(p), \beta(a), \alpha(a),$$

of length 4;

$$\alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(a), \alpha(a),$$

of length 5; and

$$\alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(r), \beta(a), \alpha(a),$$

of length 6.

It remains to construct cycles of all lengths $\ell \geq 7$.

By the induction hypothesis, $Y = \mathcal{R}_k(H)$ is vertex-pancyclic. Hence, for every integer s with

$$3 \leq s \leq M,$$

there is a cycle of length s in Y containing α.

Let

$$C : f_0, f_1, \ldots, f_{s-1}, f_0$$

be such a cycle, where $f_0 = \alpha$. By the closed lifting argument, and since every $|F_{f_i}| \geq 3$, this base cycle gives cycles through x of every length

$$2s + 1 \leq \ell \leq 3s.$$

The intervals

$$[2s + 1, 3s]$$

overlap consecutively as s ranges from 3 to M. Hence we obtain cycles through x of every length

$$7 \leq \ell \leq 3M.$$

Finally, take a Hamilton cycle

$$C : f_0, f_1, \ldots, f_{M-1}, f_0$$

of Y containing $\alpha = f_0$. Such a cycle exists by the induction hypothesis that Y is vertex-pancyclic. The closed lifting argument applied to this cycle gives cycles through x of every length in the interval

$$2M + 1 \leq \ell \leq \sum_{f \in V(Y)} |F_f| = N.$$

This interval overlaps $[7, 3M]$, since $2M + 1 \leq 3M$. Combining all cases, x lies on a cycle of every length

$$3 \leq \ell \leq N.$$

Since x was arbitrary, $\mathcal{R}_k(G)$ is vertex-pancyclic. □

Corollary 2.6. *Every nonempty d-degenerate graph has a Hamiltonian $\mathcal{R}_k(G)$ for $k \geq d + 3$; it is Hamilton-connected and vertex-pancyclic in the same range.*

3 Endpoint lifting and restoration

A colouring of a deleted graph determines the available colours at the vertices to be restored. The resulting extension graph is a fibre. In all the constructions below, assignments retain their actual coordinate values: equal assignments in two different fibres are distinct vertices, and a cross edge joins their two copies. In particular, a move of one vertex of the deleted graph may change several available lists at once.

Lemma 3.1 (Products of complete graphs)**.** *Let $s \geq 1$ and $F = \square_{j=1}^{s} K_{A_j}$, where every A_j is finite and $|A_j| \geq 2$. If all lists have size two, F has a Hamilton path between two prescribed vertices if and only if those vertices lie in opposite bipartition classes. If at least one list has size at least three, F is Hamilton-connected.*

Proof. In the binary case F is the hypercube Q_s, whose bipartition classes have equal size. The endpoint condition is therefore necessary. For sufficiency, induct on s, starting with the single edge Q_1. Choose a coordinate on which the prescribed endpoints differ and split along it. Their projections into Q_{s-1} have the same parity. Choose a projected vertex of the opposite parity; use the two induction paths to and from its copies, joined by the edge between those copies.

For the second assertion, we first show that $Y \square K_r$ is Hamilton-connected whenever Y is Hamilton-connected and $|V(Y)| \geq 3$. Induct on r, starting with $r = 1$. If the two endpoints lie in different copies of Y, order the copies starting and ending at those copies. Choose successive bridge vertices in $V(Y)$, with the first different from the first endpoint projection, consecutive bridges different, and the last different from the last endpoint projection. Such choices exist because $|V(Y)| \geq 3$: choose greedily, leaving the last bridge until the end, when it has at most two forbidden values. Prescribed-end Hamilton paths inside the copies then join into a spanning path. If both endpoints lie in one copy, take a Hamilton path between them there and choose an edge xy of it. By induction the other $r - 1$ copies have a spanning path from a copy of x to a copy of y in the same one of those copies. Insert that path in place of xy, using the two cross edges. Start with a complete-graph factor of order at least three and add the remaining factors to obtain the claim. □

Lemma 3.2 (Endpoint product-fibre lift)**.** *Let $m \geq 3$, $s \geq 1$, and let*

$$F_i = \square_{j=1}^{s} K_{A_{i,j}}, \qquad |A_{i,j}| \geq 2, \qquad 0 \leq i < m,$$

be disjoint fibres arranged along a path. Join equal assignments in consecutive fibres. At every transition assume that at least one list is unchanged and that, for every coordinate,

$$|A_{i,j} \setminus A_{i+1,j}| \leq 1, \qquad |A_{i+1,j} \setminus A_{i,j}| \leq 1.$$

If an internal fibre F_r, $1 \leq r \leq m-2$, has a factor of order at least three, then every prescribed $p \in F_0$ and $q \in F_{m-1}$ are joined by a path covering all vertices of all fibres exactly once.

Proof. Write

$$J_i = \square_{j=1}^{s} K_{A_{i,j} \cap A_{i+1,j}}$$

for the graph of common assignments. The intersections are nonempty, and an unchanged coordinate supplies an edge of J_i.

Starting at p, record the reachable exit assignments at each interface. If a fibre has a reachable entrance a, an interface edge has an admissible Hamilton-path endpoint: in a binary fibre choose its endpoint of parity opposite to a, and in a nonbinary fibre choose an endpoint different from a. Lemma 3.1 therefore shows that the reachable set never becomes empty.

At F_r, its outgoing interface J_r still contains an edge after any one assignment is removed. Indeed, if its large coordinate is unchanged, J_r contains a triangle. Otherwise that coordinate has intersection size at least two, and a different unchanged coordinate supplies a $K_2 \square K_2$. Take a reachable entrance a in F_r. Hamilton-connectedness of F_r makes every assignment in $V(J_r) \setminus \{a\}$ reachable; this set contains an entire edge.

An incoming reachable edge permits every outgoing assignment to be reached in the next fibre. In a binary fibre the two endpoints of the edge have opposite parities, so one has the parity needed for any specified outgoing assignment. In a nonbinary fibre they are distinct, so one differs from that assignment. Thus the reachable exits contain an edge at every subsequent interface. At the last fibre, one endpoint of the incoming edge can be joined to the specified q. Backtracking the finite choices supplies one Hamilton path in each fibre; the actual cross edges concatenate them into the required spanning path. □

Lemma 3.3 (Hamilton-connected independent-set restoration)**.** *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

Assume that $\mathcal{R}_k(H)$ is Hamilton-connected and that some proper k-colouring of H leaves at least three colours available at a vertex of S. If α, β are proper k-colourings of G with $\alpha|_H \neq \beta|_H$, then $\mathcal{R}_k(G)$ has an α–β Hamilton path.

Proof. Take a Hamilton path of $\mathcal{R}_k(H)$ between the two restriction colourings. For a base colouring f, the fibre is exactly

$$\square_{u \in S} K_{A_f(u)}, \qquad A_f(u) = [k] \setminus f(N_G(u)).$$

Every list is at least binary. A move at an actual vertex x of H leaves the list of at least one non-neighbour in S unchanged; every other list loses and gains at most one colour. These are the transition conditions of Lemma 3.2, even when several lists change together.

A colouring with a nonbinary fibre has at least k distinct colour permutations: fix a vertex of the nonempty H and send its colour to each of the k labels. Every resulting fibre is nonbinary. The base Hamilton path has only two endpoints, so one of these colourings is internal. Lemma 3.2 now lifts the path with the specified, actual colours of α and β on S. □

For use of this lemma, first choose a vertex w on which two prescribed colourings differ, and then choose S with $w \notin S$. This ensures different restriction colourings; the lemma does not claim to handle arbitrary endpoints in the same fibre.

Lemma 3.4 (Cyclic product-fibre lift)**.** *Let $m \geq 3$ disjoint fibres $F_i = \square_{j=1}^{s} K_{A_{i,j}}$, with $s \geq 1$ and $|A_{i,j}| \geq 2$, be cyclically ordered. Join equal assignments in consecutive fibres and define $J_i = \square_j K_{A_{i,j} \cap A_{i+1,j}}$, with indices modulo m. Suppose every J_i contains an edge, F_0 has a factor of order at least three, and there are distinct edges $e_0 \in E(J_0)$ and $e_{m-1} \in E(J_{m-1})$ (compared as actual assignment pairs). Then the layered graph has a Hamilton cycle.*

Proof. Choose an edge e_i in every remaining interface, and choose $z_0 \in V(e_0) \setminus V(e_{m-1})$. For $1 \leq i < m$, there is a bijection $\phi_i : V(e_{i-1}) \to V(e_i)$ such that each $z, \phi_i(z)$ is an admissible pair of Hamilton-path endpoints in F_i. In a binary fibre match opposite parities. Otherwise use a bijection with no fixed assignment, which always exists between two two-element sets: interchange equal sets, send a common member to the other target member when there is exactly one common member, and use either bijection when the sets are disjoint.

Put $z_i = \phi_i(z_{i-1})$. Traverse F_i from z_{i-1} to z_i for $1 \leq i < m$. Since $z_{m-1} \in V(e_{m-1})$, $z_{m-1} \neq z_0$, so the nonbinary F_0 has a Hamilton path from z_{m-1} to z_0. Join the fibre paths by the cross edges at the assignments z_i. They cover every fibre once and form one cycle. □

Lemma 3.5 (Independent-set restoration)**.** *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

If $\mathcal{R}_k(H)$ is Hamiltonian and some proper k-colouring of H leaves at least three colours available at a vertex of S, then $\mathcal{R}_k(G)$ is Hamiltonian.

Proof. Take a Hamilton cycle of the base and use the same product fibres as in Lemma 3.3. Every interface contains an edge. Index a nonbinary fibre as F_0 and choose a large coordinate in it. At either adjacent interface, that coordinate supplies a triangle if unchanged, or a $K_2 \square K_2$ together with a different unchanged coordinate if it changes. Thus both adjacent interfaces have at least two edges, and distinct edges e_0, e_{m-1} can be chosen. Apply Lemma 3.4. □

Lemma 3.6 (Single-vertex Hamilton-connected lift)**.** *Let $H = G - v \neq \varnothing$, $k \geq 4$, and $k \geq d_G(v) + 3$. Assume that v has a non-neighbour in G, $\mathcal{R}_k(H)$ is Hamilton-connected, and $\delta(\mathcal{R}_k(H)) > d_G(v)$. Then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. For a base colouring f, write $A_f = [k] \setminus f(N_G(v))$. The fibre is the clique K_{A_f}, $|A_f| \geq 3$, and adjacent base fibres have intersections of size at least two.

Let w be a non-neighbour of v. Any base path covering all states, or all but one state, has a step changing w. Indeed, global colour permutations show that w takes all k colours among the base states; omitting one state leaves at least $k - 1 \geq 3$ colours. At a step changing w, the available list for v is unchanged, so its intersection has size at least three. Consequently such a base path can be lifted with arbitrary prescribed first and last fibre colours. To see this, choose bridge colours in the successive intersections, requiring consecutive bridge colours to differ and avoiding the prescribed endpoint colour at either end. All intersection lists have size at least two and one has size at least three. Greedily choose from the two ends towards a three-element list, and choose at that list last while avoiding its at most two forbidden colours. Then traverse each clique between its distinct entrance and exit colours.

If the prescribed states lie in different fibres, lift the base Hamilton path between their restrictions. Suppose instead that both lie over f, with distinct colours a, b. A Hamilton-connected graph with at least three vertices has a Hamilton cycle through every edge: close a Hamilton path between the ends of the edge by that edge. Here the base has at least k states.

If $|A_f| \geq 4$, choose a base Hamilton cycle through f, whose neighbours on the cycle are g, h. Both intersections with A_f have size at least three. Choose $p \in A_f \cap A_g \setminus \{a, b\}$ and $q \in A_f \cap A_h \setminus \{a, p\}$. Split the root clique into disjoint paths a–p and q–b covering it; the second may be a single vertex if $q = b$. The base cycle minus f is a path covering all other states, and the preceding lifting observation traverses those fibres with endpoint colours p, q. The two cross edges and the two root paths give the desired spanning path.

If $A_f = \{a, b, c\}$, at most $d_G(v)$ base moves from f make c unavailable: for each neighbour of v, only its move to colour c can do so. The state-degree assumption gives a base edge fg preserving c. Choose a base Hamilton cycle containing fg and write h for the other neighbour of f on it. Select $q \in (A_f \cap A_h) \setminus \{c\}$, so $q = a$ or b. If $q = b$, start with the root edge ac, cross at colour c, lift the path through all other base states, and return to the root at colour b. If $q = a$, carry out the same construction with the two endpoints interchanged and reverse the resulting path. This covers the three root states and all other fibres exactly once. □

A Hamilton-connected graph of order at least four has minimum degree at least three. A vertex of degree two would, with its neighbours chosen as the two endpoints, force a spanning path to have only three vertices; degrees zero and one are excluded as well. Every nonempty colourable H has at least k states in $\mathcal{R}_k(H)$, by colour permutation. These observations will supply the state-degree condition in the low-degree inductions.

4 Hamilton-connected planar classes

4.1 All planar graphs

We first establish the marked structural statement needed to retain a coordinate on which the prescribed endpoint colourings differ.

Lemma 4.1 (Separated degree-five vertices with a marked vertex)**.** *If G is planar and $\delta(G) \geq 5$, then, for any $w \in V(G)$, there are two degree-five vertices different from w whose distance is at least three. Vertices in different components have infinite distance.*

Proof. Extend G, without adding vertices, to a simple plane triangulation T. A degree-five vertex in T still has degree five in G, and a pair at distance at least three in T has that property in G. Moreover $\delta(T) \geq 5$. It suffices to prove the assertion in T. Euler's formula gives at least twelve degree-five vertices.

For a degree-five vertex x, put $W = N_T[x]$, $U = V(T) \setminus W$, $r = |U|$, and let B be the vertices at distance exactly two from x; write $b = |B|$. The five neighbours of x form a cycle in their cyclic order, so $e_T(W) \geq 10$. Contract the connected W and simplify parallel edges. The resulting plane graph has $r + 1$ vertices and $e_T(U) + b$ edges. As $|V(T)| \geq 12$, the planar edge bound gives $e_T(U) + b \leq 3r - 3$. Since $e(T) = 3r + 12$,

$$\sum_{z \in U} d_T(z) = e_T(U) + e(T) - e_T(W) \leq 6r - b - 1.$$

Let F_x be the degree-five vertices at distance at least three from x. The same sum is exactly

$$6r - b - |F_x| + \sum_{z \in B} (d_T(z) - 5) + \sum_{\substack{z \in U \setminus B \\ d_T(z) \geq 6}} (d_T(z) - 6).$$

It follows that $|F_x| \geq 1$, and if $|F_x| = 1$, every vertex of B has degree five.

Suppose every pair of degree-five vertices at distance at least three contains w, and let L be all degree-five vertices except w. For every $x \in L$, the nonempty F_x must be $\{w\}$. Hence w is at distance at least three from L, and every vertex at distance exactly two from x belongs to L. Take a shortest path from w to L, ending at x. The vertex two steps before x is also in L and is closer to w, a contradiction. □

Theorem 4.2 (Planar Hamilton-connectedness)**.** *For every nonempty planar graph G and every $k \geq 7$, $\mathcal{R}_k(G)$ is Hamilton-connected. Consequently $h(\text{Planar}) = \text{hc}(\text{Planar}) = 7$.*

Proof. Prove the seven-colour assertion by induction on $|V(G)|$, allowing disconnected graphs throughout. If $|V(G)| \leq 5$, the graph is four-degenerate and Theorem 2.4 applies. For $|V(G)| \geq 6$, a vertex v of degree at most four has a non-neighbour. The smaller $H = G - v$ is Hamilton-connected by induction. For every seven-colouring f of H, which has at least five vertices,

$$d_{\mathcal{R}_7(H)}(f) = \sum_{x \in V(H)} (6 - |f(N_H(x))|) \geq 6|V(H)| - 2|E(H)| \geq 12 > d_G(v).$$

Lemma 3.6 restores v.

Otherwise $\delta(G) \geq 5$. For prescribed distinct colourings α, β, choose an actual vertex w with $\alpha(w) \neq \beta(w)$. Lemma 4.1 supplies $S = \{u, v\}$, avoiding w, with $d(u) = d(v) = 5$ and $d_G(u, v) \geq 3$. Thus S is independent, no outside vertex is adjacent to both members, and $H = G - S$ is smaller and nonempty. Choose two nonadjacent neighbours x, y of u; they exist since its five neighbours cannot induce a planar K_5. Properly six-colour the planar $H - \{x, y\}$, and give both x, y colour seven. This is a proper colouring of H in which the neighbours of u use at most four colours, leaving at least three available. The restriction colourings differ at w, so Lemma 3.3 gives an α–β Hamilton path.

For $k \geq 8$, planar five-degeneracy and Theorem 2.4 suffice. For sharpness, colour the six antipodal vertex pairs of the icosahedron with six different colours. Each vertex sees all other five colours, so this is a frozen proper six-colouring; its recolouring graph has more than one state and is disconnected. Thus both thresholds are seven. □

Planar five-degeneracy and Theorem 2.5 also give vertex-pancyclicity for $k \geq 8$.

4.2 Triangle-free and bipartite planar graphs

The critical five-colour step needs an adjacent-pair fibre, which is not a product of complete graphs. We give its finite input separately from the arbitrary-size lifting argument.

For $A, B \subseteq \{0,1,2,3,4\}$, put

$$F(A,B) = \{(a,b) \in A \times B : a \neq b\},$$

with adjacency given by changing one coordinate. When $|A|, |B|$ are three or four, exchange the coordinates if necessary so that $|A| \leq |B|$. There are exactly seven types $(|A|, |B|, |A \cap B|)$.

Lemma 4.3 (The seven adjacent-pair types)**.** *The five types* $331, 342, 343, 443, 444$ *are Hamilton-connected. The type* 333 *is a* 6*-cycle, so its feasible Hamilton-path endpoint pairs are exactly its six edges. For the normalized type* 332*,* $A = \{0,1,2\}$ *and* $B = \{0,1,3\}$*, all distinct endpoint pairs are feasible except* $\{03, 21\}$ *and* $\{13, 20\}$*, where* ab *denotes* (a,b)*.*

Proof. The explicit words in Appendix B prove every positive assertion: each word lists the complete state set once, with consecutive words differing in one coordinate. Apply all colour permutations preserving the listed pair of lists, the coordinate exchange when it preserves the type, and path reversal. The resulting endpoint pairs have the counts in Table B.1, covering all claimed positive pairs.

For 332, the two excluded pairs are the neighbour pairs of the degree-two states 01 and 10, respectively. Choosing such a pair as spanning-path ends would force the entire path to have only three vertices. For 333, inspection gives the stated 6-cycle. These supply exact negative witnesses, rather than search failures. □

Lemma 4.4 (Adjacent-pair interfaces and reset)**.** *Arrange full fibres* $F(A_i, B_i)$ *along a base path. Each list has size three or four in a fixed five-colour palette. At each step either both lists stay unchanged, or one stays unchanged and the other is replaced by a three-element list differing in one colour, or a three-element list and a containing four-element list are interchanged. Join the copies of equal assignments at consecutive layers. If an internal layer has* $|A_r| = |B_r| = 4$*, there is a spanning path between any prescribed first-layer and last-layer states.*

Proof. First consider one allowed interface between $F(A,B)$ and $F(A',B')$, and write J for their common assignment set. It has at least four states. For every $p \in F(A,B)$, there is a $q \in J$ such that the first fibre has a p–q Hamilton path. In a Hamilton-connected type choose any $q \neq p$. In type 332, each p forbids only itself and at most one additional endpoint, so again a choice exists. In type 333, J is either its whole 6-cycle or that cycle minus a row or column of two adjacent states. A vertex's two cycle neighbours are not adjacent to one another, so at least one remains in J. The Hamilton path obtained by deleting their connecting cycle edge has the desired endpoints. The same property holds at the reverse interface.

Use this property to propagate nonempty sets of reachable exits from the prescribed first state. The internal 44 layer is Hamilton-connected, and its outgoing interface has at least nine states. Any reachable entrance p can therefore reach all interface states except possibly p, yielding at least two distinct reachable exits. The next layer still has one four-element list and is one of $342, 343, 443, 444$, hence is Hamilton-connected. Two distinct reachable entrances let every state of that layer be an exit. If it is the last layer, this includes the prescribed endpoint. Otherwise its entire outgoing interface is reachable.

Thereafter this property propagates: for a specified exit q in the current layer, apply the one-interface property in reverse to find a Hamilton-path entrance in its fully reachable incoming

interface. The last prescribed endpoint is therefore reachable. Backtracking gives one Hamilton path in every full fibre, joined by actual cross edges, with no repetition or omission. □

Theorem 4.5 (Triangle-free planar Hamilton-connectedness)**.** *If G is a nonempty triangle-free planar graph, then $\mathcal{R}_k(G)$ is Hamilton-connected for every $k \geq 5$. For the class of nonempty triangle-free planar graphs, $h = \mathrm{hc} = 5$.*

Proof. For $k \geq 6$, use three-degeneracy and Theorem 2.4. For $k = 5$, induct on the order of G, permitting disconnected graphs. Orders at most three are handled by two-degeneracy and the same theorem. At larger orders a vertex v of degree at most two has a non-neighbour. The induction hypothesis gives a Hamilton-connected base of at least five states, thus of minimum degree at least three; Lemma 3.6 applies.

Assume now $\delta(G) \geq 3$. Let L be its degree-three vertices and $s = |L|$. The triangle-free planar edge bound gives

$$s \geq 8 + \sum_{d_G(z) \geq 4} (d_G(z) - 4).$$

Given distinct α, β, choose w on which they differ. There are two exhaustive cases.

If $L \setminus \{w\}$ contains adjacent u, v, denote their other neighbours by u_1, u_2 and v_1, v_2. The four actual outside vertices are distinct: a common neighbour of u and v would form a triangle. In $H = G - \{u, v\}$ the restriction endpoints still differ at w, so induction gives a base Hamilton path between them. Its fibre over f is

$$F(A_f, B_f), \qquad A_f = [5] \setminus \{f(u_1), f(u_2)\}, \quad B_f = [5] \setminus \{f(v_1), f(v_2)\}.$$

The lists have size three or four. One actual base move affects at most one of them and gives precisely an interface allowed in Lemma 4.4. Edges among the four outside vertices restrict the base colourings, but do not alter this statement.

There is a base colouring with both lists four-element: use Grötzsch's theorem [6] to three-colour $H - \{u_1, u_2, v_1, v_2\}$, then give u_1, u_2 colour four and v_1, v_2 colour five. Each pair is independent by triangle-freeness, and edges across the pairs have distinct colours. At least five global colour permutations give distinct double-four-list base states. Since the Hamilton path has only two endpoints, one is internal. Apply Lemma 4.4 with the specified local endpoint states to complete the restoration.

Otherwise $S = L \setminus \{w\}$ is independent and has at least $s - 1 \geq 7$ vertices. For any outside vertex z of degree at least four, the displayed Euler budget implies $d_G(z) \leq s - 4 < |S|$. The only possible degree-three outside vertex is w, whose degree is also less than $|S|$. Thus no outside vertex sees all of S. Choose $u \in S$ and two of its neighbours x, y. They are nonadjacent. Three-colour $H - \{x, y\}$ by Grötzsch's theorem and give both x, y colour five. At most two colours then occur on the three neighbours of u, so u has at least three available colours. All conditions of Lemma 3.3 hold, and w is retained; it gives the desired endpoint path.

For sharpness, on the cube Q_3 view the colours as $\mathbb{Z}_2^2$ and set

$$f(x, y, z) = x(1, 0) + y(0, 1) + z(1, 1).$$

The three neighbours of every vertex have all three other colours, so the colouring is frozen and the complete four-colour recolouring graph is disconnected. The upper and lower bounds give both thresholds. □

Corollary 4.6. *The class of nonempty bipartite planar graphs has $h = \mathrm{hc} = 5$. For every graph in this class, $\mathcal{R}_k(G)$ is vertex-pancyclic for $k \geq 6$.*

Proof. Bipartite planar graphs are triangle-free, and the cube belongs to this class. The vertex-pancyclic assertion follows from three-degeneracy and Theorem 2.5. □

The same vertex-pancyclic bound $k \geq 6$ holds for all triangle-free planar graphs. This conclusion uses the degenerate theorem separately from Hamilton-connectedness.

4.3 Girth at least six and forests

Lemma 4.7 (Separated degree-two vertices with a marked vertex)**.** *If G is planar, $g(G) \geq 6$, and $\delta(G) \geq 2$, then for any $w \in V(G)$ there are two degree-two vertices different from w whose distance is at least three.*

Proof. Every component has a cycle, and the girth-six Euler bound gives at least six degree-two vertices per component. If there are at least two components, choose such a vertex different from w in each of two components. Suppose G is connected and let n_2 be its number of degree-two vertices. From $2|E(G)| \leq 3|V(G)| - 6$,

$$n_2 \geq 6 + \sum_{d_G(z) \geq 3} (d_G(z) - 3).$$

Let L consist of all degree-two vertices except w; then $|L| \geq n_2 - 1 \geq 5$. Suppose all pairs in L have distance at most two.

If $u, v \in L$ are adjacent, let a, b be their respective other neighbours. The absence of cycles of length at most five implies that the intersection of the two closed distance-two balls is exactly $\{u, v, a, b\}$, and that $d_G(a, b) = 3$. Thus at most three of these vertices can lie in L, a contradiction.

Otherwise L is independent. Its two-element neighbourhoods are pairwise intersecting; they are distinct because there is no four-cycle. At least four distinct pairwise intersecting two-element sets have a common element: if $\{a, b\}, \{a, c\}$ occur and a third set omits a, it must be $\{b, c\}$, after which no fourth distinct set is possible. Hence some vertex z is adjacent to all of L. Then $d_G(z) \geq n_2 - 1 \geq 5$, whereas the Euler inequality gives

$$n_2 \geq 6 + d_G(z) - 3 \geq n_2 + 2,$$

a contradiction. □

Theorem 4.8 (Girth at least six)**.** *For every finite integer $g \geq 6$, the class of nonempty planar graphs of girth at least g has $h = \mathrm{hc} = 4$. Every graph G in this class has Hamilton-connected $\mathcal{R}_k(G)$ for $k \geq 4$, and vertex-pancyclic $\mathcal{R}_k(G)$ for $k \geq 5$.*

Proof. For $k \geq 5$, use two-degeneracy and Theorems 2.4 and 2.5. At $k = 4$, induct on the order, starting with orders at most two, which are forests and covered by the one-degenerate theorem. If $|V(G)| \geq 3$ and $d_G(v) \leq 1$, then v has a non-neighbour. The smaller nonempty base is Hamilton-connected by induction and has at least four states, hence minimum degree at least three. Restore v using Lemma 3.6.

Otherwise apply Lemma 4.7 to a vertex w at which the prescribed endpoint colourings differ. It gives separated degree-two vertices u, v, both different from w. Set $S = \{u, v\}$ and $H = G - S$. The set S is independent, no outside vertex sees both its members, and H is nonempty. The neighbours x, y of u are nonadjacent. The two-degenerate $H - \{x, y\}$ can be three-coloured; giving x, y colour four leaves three colours available at u. Lemma 3.3 now restores a Hamilton path between the actual prescribed states.

For the lower bound, take a cycle of length at least g divisible by three and repeat the colours $0, 1, 2$ around it. This is a frozen proper three-colouring, so its complete three-colour recolouring graph is disconnected. Both class thresholds are therefore four. □

Theorem 4.9 (Forests)**.** *The class of nonempty forests has* $h = \mathrm{hc} = 4$*. Every such forest* F *has Hamilton-connected and vertex-pancyclic* $\mathcal{R}_k(F)$ *for* $k \geq 4$*.*

Proof. The upper bounds follow from one-degeneracy and Theorems 2.4 and 2.5. For the Hamiltonian lower bound, $\mathcal{R}_3(P_3)$ has twelve states. Its six states using three distinct colours have degree two, so every edge incident with them is forced in a spanning cycle. For each of the three middle-vertex colours b, these forced edges form a 4-cycle

$$(a, b, a), (c, b, a), (c, b, c), (a, b, c), \qquad \{a, b, c\} = \{0, 1, 2\}.$$

The three disjoint forced cycles prevent a Hamilton cycle. For the Hamilton-connected lower bound, the subgraph $\mathcal{R}_3(K_2) = C_6$ is already not Hamilton-connected. □

4.4 The single graph C_5

Theorem 4.10 (The five-cycle)**.** *The individual graph* C_5 *has* $h(C_5) = \mathrm{hc}(C_5) = 4$*. Its four-colour recolouring graph is Hamilton-connected, its three-colour recolouring graph is the disjoint union of two* 15*-cycles, and* $\mathcal{R}_k(C_5)$ *is Hamilton-connected and vertex-pancyclic for* $k \geq 5$*.*

Proof. The four-colour assertion is a complete finite certificate. The state set consists of the 240 proper words of length five over four colour labels. Under colour permutation and the dihedral symmetries of the cycle, every state is equivalent to one of 01012 and 01023. For each of these two initial states the certificate records a Hamilton path to each of the other 239 states. Its 478 paths each contain all 240 states exactly once, have the required endpoints, and have a single-coordinate change at each of the 114242 steps. The group actions preserve properness and adjacency, so these paths cover every ordered pair of distinct endpoint states. This establishes precisely the complete finite graph $\mathcal{R}_4(C_5)$.

Direct enumeration at three colours gives 30 proper states, all of degree two, with two components of order fifteen. Hence that graph is two 15-cycles and is not Hamiltonian or Hamilton-connected. Finally, C_5 is two-degenerate, so the separate assertions for $k \geq 5$ follow from Theorems 2.4 and 2.5. The finite certificate makes no claim about other girth-five graphs. □

5 Hamiltonian four-colour subclasses at girth five

The results in this section are Hamiltonian results. They do not assert Hamilton-connectedness at four colours. The local models are defined below and their literal certificates are recorded in the appendices. Their finite parts classify labelled lists; the inductions are valid for an arbitrary number of vertices and are not extrapolations from finite state spaces.

Lemma 5.1 (Subdivided-star lift)**.** *Let* $S = \{u_1, \ldots, u_s\}$ *be independent,* $s \geq 3$*, and suppose* $N(u_i) = \{w, x_i\}$*, where* $N(w) = S$*,* $w \notin S$*, and the* x_i *are distinct and belong to* $J = G - (S \cup \{w\})$*. If* $\mathcal{R}_4(J)$ *is Hamiltonian and some proper colouring of* J *gives the same colour to two distinct* x_i*, then* $\mathcal{R}_4(G)$ *is Hamiltonian.*

Proof. Take a Hamilton cycle C of $\mathcal{R}_4(J)$. Choose three distinct edges of C. At each edge at most one x_i changes, so an unchanged x_j supplies a colour a_r at the rth chosen edge. Choose a tree on the four colours whose three edges contain a_1, a_2, a_3, respectively (a star works when there is one repeated label; if the labels are all distinct, use the star at the fourth colour). Make four copies of C, indexed by the colour of w, and splice the corresponding copies along the three selected edges

according to that colour tree. The result is a Hamilton cycle B of $\mathcal{R}_4(G-S)$ with the following control: whenever w changes from a to b along B, some x_j has colour a or b.

Over a base colouring f the u_i-lists are

$$A_i(f) = [4] \setminus \{f(w), f(x_i)\}.$$

They have size at least two. At a move in J, one list changes and another coordinate is unchanged. At a move of w, all intersections are $[4] \setminus \{a, b, f(x_i)\}$, and the control property leaves a coordinate with a two-element intersection. Thus the product-fibre splice applies at every step. At the base colouring with $f(x_p) = f(x_q)$, give w that same colour; the two corresponding lists have size three, so both interfaces at this fibre contain two distinct splice edges. The resulting cycle traverses all extensions and proves the lemma. □

Lemma 5.2 (Five-face identity)**.** *Let C be a connected plane graph of girth at least five and minimum degree at least two. Let p be the number of 5-faces, $Q = \sum_{|F|\geq 7}(|F|-6)$, $s = |\{v : d(v) = 2\}|$, and $D = \sum_v (d(v)-3)$ over vertices of degree at least three. Then*

$$p = 12 + Q + 2D - 2s.$$

Every 5-face is a simple 5-cycle, and distinct faces give distinct cycles unless $C = C_5$.

Proof. A facial walk has no immediate reversal and contains a cycle, hence has length at least five. Counting face lengths gives $2e = 6f - p + Q$, while the degree sum is $2e = 3n - s + D$; with $n - e + f = 2$, subtraction yields the identity. A repeated vertex on a facial walk of length five would give a shorter cycle. If one 5-cycle bounds two faces, all its vertices have degree two and connectedness gives $C = C_5$. □

Corollary 5.3 (Two-degeneracy from seven pentagons)**.** *A planar graph of girth at least five with at most seven distinct 5-cycles is two-degenerate.*

Proof. Otherwise a connected subgraph has minimum degree at least three. In its plane embedding the identity has $s = 0$, and therefore at least twelve 5-faces, which are distinct 5-cycles. This contradicts the hypothesis. □

Theorem 5.4 (At most seven distinct five-cycles)**.** *Let G be a nonempty planar graph with girth at least five and at most seven distinct 5-cycles. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$; therefore the Hamiltonian threshold of this class is four. No four-colour Hamilton-connected conclusion is claimed.*

Proof. For $k = 4$ use induction on $|V(G)|$, allowing disconnected graphs. The one-vertex graph gives K_4. A vertex of degree at most one is restored by Lemma 2.2. If two degree-two vertices are adjacent, use Lemma 5.5. Otherwise the degree-two vertices form an independent set S. In a component containing a cycle, Lemma 5.2 and the bound $p \leq 7$ give $|S| \geq 3$; a tree component has a leaf and was already reduced.

If no vertex is adjacent to all of S, every vertex outside S has at most $|S| - 1$ neighbours in S. Choose $u \in S$ with neighbours x, y. They are nonadjacent by the girth condition. Three-colour $G - S - \{x, y\}$ greedily using Corollary 5.3, and colour both x, y with a fourth colour. This gives a base colouring with three colours available at u. Lemma 3.5 restores S.

Suppose that a vertex w is adjacent to all of S. In the component under consideration write $d(w) = |S| + t$. The identity and $D \geq d(w) - 3$ give $p \geq 6 + Q + 2t$, so $p \leq 7$ forces $t = 0$ and $N(w) = S$. Write $N(u_i) = \{w, x_i\}$. The x_i are distinct, and two are nonadjacent since $|S| \geq 3$

and the graph has no triangle. Put $J = G - (S \cup \{w\})$. Greedy three-colouring of $J - \{x_p, x_q\}$ followed by assigning the fourth colour to x_p, x_q supplies the hypothesis of Lemma 5.1; induction on J completes this case. For $k \geq 5$, apply the degenerate Hamiltonian theorem to the corollary above (and use the one-vertex lift at each deletion); this is a separate argument and assumes no monotonicity in k. □

Lemma 5.5 (Adjacent degree-two restoration)**.** *Let u, v be adjacent degree-two vertices, with $N(u) = \{v, x\}$ and $N(v) = \{u, y\}$, where $x \neq y$. If $\mathcal{R}_4(G - \{u, v\})$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. For a base colouring f, the extension fibre is $F(A, B)$ with $A = [4]\backslash\{f(x)\}$ and $B = [4]\backslash\{f(y)\}$. If $A = B = \{a, b, c\}$, use

$$(ab, ac, bc, ba, ca, cb).$$

If $A = \{a, b, c\}$ and $B = \{a, b, d\}$, use either

$$(ab, ad, bd, ba, ca, cd, cb), \qquad (ab, ad, cd, bd, ba, ca, cb).$$

A one-colour list change has two common labelled internal edges between any old and new selected cycles; in the equal case use the two unchanged rows, and in the unequal case inspect the two displayed words. Cut an edge of a base Hamilton cycle and splice successive fibres along these two common edges. Every splice joins two disjoint cycles and the fibre states are partitioned, so the resulting cycle is spanning. □

5.1 Two degree-two reductions

Lemma 5.6 (Separated degree-two vertices)**.** *Let G be planar of girth at least five and let u, v be nonadjacent vertices of degree two with $N(u) \cap N(v) = \varnothing$. If $H = G - \{u, v\}$ has Hamiltonian $\mathcal{R}_4(H)$, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Set $S = \{u, v\}$. Every vertex of H has at most one neighbour in S. Let x, y be the neighbours of u; they are distinct and nonadjacent. The planar triangle-free graph $H - \{x, y\}$ admits a proper three-colouring by Grötzsch's theorem [6] (the empty graph has its empty colouring). Use colours $0, 1, 2$ there and give both x, y colour 3. This is a proper colouring of H leaving three colours available at u. The hypotheses of Lemma 3.5 hold with $k = 4$ and $|S| = 2$, so that lemma supplies the required Hamilton cycle. □

5.2 Restoring a $(2, 3, 2)$ path

The following reduction uses only a Hamilton cycle of the deleted graph. It processes an entire constant-boundary run together with its two neighbouring nonconstant fibres, retaining the edges needed for the next splice. The three external neighbours remain distinct actual vertices.

Proposition 5.7. *If $F = \mathcal{R}_3(P_3)$, then $F \,\square\, P_\ell$ is Hamiltonian for every $\ell \geq 2$.*

Proof. Partition F into its three centre-colour squares Q_b. Order each square as (q_0, q_1, q_2, q_3), where q_0, q_2 are its monochromatic-endpoint vertices. On $Q_b \square P_2$ use

$$(0, q_0), (1, q_0), (1, q_1), (0, q_1), (0, q_2), (1, q_2), (1, q_3), (0, q_3).$$

This cube cycle contains all four edges between layers 0 and 1. For larger ℓ, append each subsequent square by an ordinary edge splice using an internal edge of the newest layer. The initial newest layer

has two internal edges; each appended layer retains three. This leaves all four original interlayer edges intact.

The three centre-colour squares of F are connected by three edges between monochromatic-endpoint vertices, one for each pair of centre colours. Choose a tree of two of these edges. Each chosen edge xy produces a square with vertices $(0,x),(1,x),(1,y),(0,y)$. Delete its two interlayer edges and add its two copies of xy, joining the corresponding strip cycles. The two selected interlayer edges at the middle strip are distinct. Two such splices give a single spanning cycle, for arbitrary $\ell \geq 2$. □

Let x, y, z be a path of degrees $2, 3, 2$. Its external neighbours, in that order, are p, q, r. Assume they are distinct and outside the path. For a bottom colouring with boundary $t = (f(p), f(q), f(r))$, write

$$B_t = \{(u,v,w) : u \neq v,\ v \neq w,\ u \neq t_0,\ v \neq t_1,\ w \neq t_2\}, \qquad u,v,w \in \{0,1,2,3\}.$$

A boundary is *bad* if all three entries agree, and *good* otherwise. Up to colour permutations and path reversal the good types are $001, 010, 012$. Their orders are $6+4+4=14$, $9+4+4=17$, and $6+6+4=16$, respectively. The bad type 000 has order 12.

5.3 Good cycles with safe connecting edges

The cycles D_E, D_M, D_A for the normalized boundaries $001, 010, 012$ are specified in (C.1) in Appendix C; the coordinates are uvw. Allow all their colour images and path reversals. Legality, cyclic adjacency, distinctness, and the counts above verify that each is a Hamilton cycle of its fibre.

For a bad boundary aaa, partition B_{aaa} into squares Q_c indexed by $c \neq a$, with fixed middle coordinate c. A good neighbour of aaa is obtained by changing position j from a to $b \neq a$. If j is an end position, then for each $c \notin \{a,b\}$ there is exactly one edge common to the good fibre and Q_c. Let P_t consist of these two edges. For middle-change or all-distinct boundaries set $P_t = \varnothing$. A good-cycle edge outside P_t is called *safe*. Directly from (C.1), every allowed cycle has the following properties:

1. For an end change, it contains the unique edge in each Q_c, $c \notin \{a,b\}$, and two opposite edges of Q_b.
2. For a middle change it has no vertex in Q_b and contains three edges of each of the other two squares.

For $t = 001$, the two special edges are 123–323 and 132–232; the two edges in Q_1 are 313–213 and 212–312. For $t = 010$, the retained three-edge paths are $323, 123, 121, 321$ and $231, 232, 132, 131$. These descriptions also prove the assertions for every symmetry image.

Lemma 5.8 (Safe common edges)**.** *For adjacent good boundaries t, s, every allowed cycle at t and every allowed cycle at s have at least two common edges outside $P_t \cup P_s$.*

Proof. The edge numbering and target-column conventions are specified in Appendix C. Table C.1 supplies two safe common source-edge indices for every normalized source and allowed target cycle.

To prove completeness, order the types as $E < M < A$ and reverse the boundary transition so that its source has no larger type than its target. Normalize its chosen source cycle to (C.1) by a colour permutation and possibly reversal. There are eight eligible targets from 001, six from 010, and three from 012, exactly the seventeen rows of Table C.1. The column rules exhaust every target cycle. Each of the thirty-seven entries is verified by substituting its stated colour map and reversal

in (C.1), and excluding the two explicitly described special edges where applicable. Thus the table is a complete finite analytic certificate. The transformations preserve special edges, completing the proof. □

5.4 A bad block with prescribed safe edges left intact

Lemma 5.9 (Bad-block lifting)**.** *Consider consecutive fibres with boundaries*

$$t, \underbrace{aaa, \ldots, aaa}_{\ell \text{ times}}, s, \qquad \ell \geq 1,$$

where t, s are good and each differs from aaa in one coordinate. When $\ell = 1$ suppose $t \neq s$. Choose any allowed good cycles at the two ends, and reserve at most one safe edge in each. The layered graph on these fibres has a Hamilton cycle retaining both reserved edges. Only consecutive-layer matching edges are used.

Proof. Call a common edge with Q_c *available* if it belongs to the chosen good cycle and is not its reserved edge. An end-change good cycle has available edges in all three squares: the two singleton edges are special, hence cannot have been reserved, and at least one of its two Q_b edges remains. A middle-change good cycle has at least two available edges in each of its two supported squares.

First suppose $\ell = 1$. Start with five disjoint cycles: the two good cycles and the three bad squares. Choose a bridge square with two different available edges, one for each good cycle; attach the other two squares as leaves to good cycles. The following cases guarantee such a choice. Denote the changed colours by b_L, b_R.

1. Both changes are at ends. Choose $c \notin \{a, b_L, b_R\}$. Both singleton edges in Q_c are available. They are different: if the changed end is the same, $t \neq s$ implies $b_L \neq b_R$, and the edges are opposite in Q_c; if the changed ends differ, the edges change different endpoint coordinates. Either good cycle can receive either remaining square.

2. Both changes are at the middle. Here $b_L \neq b_R$. Use the remaining colour c for the bridge square. Each side has at least two available edges there, so different edges can be selected. Attach Q_{b_L} to the right and Q_{b_R} to the left.

3. One change is at the middle, with colour b_M, and the other at an end, with colour b_E. Choose $c \notin \{a, b_M, b_E\}$. The end's singleton edge in Q_c is available, and the middle side has an available edge different from it. Attach Q_{b_M} to the end-change cycle, and the last square to the middle-change cycle. This also covers $b_M = b_E$: the end-change cycle retains at least one of its two Q_{b_M} edges.

The four selected incidences form a tree on the five initial cycles. Cuts on a good cycle occur at different middle colours, hence are distinct; the two bridge-square cuts were chosen distinct. Along each tree edge remove the corresponding internal edge from each cycle and add the two matching edges. Successively adding a new tree component merges two cycles into one. All reserved edges survive.

Now let $\ell \geq 2$. Each good end supports at least two squares, so choose an available left edge in Q_c and an available right edge in Q_d with $c \neq d$. Build the three square-strip cycles as in Proposition 5.7. The cube formula there permits either perfect matching of a square to be used in the first layer, by rotating its cyclic order; the last layer of a two-layer strip uses the other matching. Consequently its first layer can retain any specified square edge. For a two-layer strip the same is true at its last layer; our two requirements are on different strips and are independent.

If $\ell \geq 3$, a last-layer requirement is met by appending each new square with a splice edge different from the required edge. There are initially two choices and subsequently three in the current last layer, so this is always possible. The new last square retains all edges except the splice edge. First-layer horizontal edges and all four original vertical edges survive these extensions.

Join the three strips using the two vertical-edge switches in Proposition 5.7; these do not affect either required horizontal edge. Finally join the left and right good cycles to their selected strip edges by two ordinary edge splices. The result is one Hamilton cycle. The reserved edges were never cut. This proves the claim for every length, rather than just a bounded set of runs. □

5.5 The complete path-lifting induction

Lemma 5.10 ($(2,3,2)$-path lifting)**.** *Suppose the three external neighbours of a degree-$(2,3,2)$ path are distinct and outside it. If the reconfiguration graph of the deleted graph is Hamiltonian at four colours, then so is $\mathcal{R}_4(G)$.*

Proof. Fix an arbitrary bottom Hamilton cycle. There must be a good vertex on it. Otherwise every boundary is constant; consecutive constant triples can only be equal, as at most one coordinate changes. The external colour would therefore be fixed throughout the cycle. A global colour permutation of a proper colouring supplies another proper colouring with a different external colour, a contradiction to Hamilton coverage. Thus even the startup uses only Hamiltonicity.

Every good bottom vertex has at most one bad neighbour in the entire bottom reconfiguration graph: a two-colour boundary has exactly one possible coordinate change making it constant, and an all-distinct boundary has none. The full neighbouring colouring, if proper, is uniquely determined by that change. Hence no good vertex on the Hamilton cycle can lie between two bad vertices. In particular there is an edge joining two good vertices.

Cut such an edge. In the resulting bottom Hamilton path, group each maximal bad run together with its two neighbouring good vertices; make each remaining good vertex a singleton block. These blocks are disjoint and partition the path. A bad run has constant boundary, and its two neighbours are distinct. If the run has length one their boundaries are different: otherwise both neighbours are obtained from the intervening colouring by the same one-vertex recolouring and would be the same bottom vertex.

For each labelled good boundary fix one allowed cycle once and for all. Consecutive blocks meet at a good–good boundary transition. If the boundary changes, Lemma 5.8 provides two common safe edges. If it stays unchanged, the fixed cycles coincide and have at least $14 - 2$ safe edges. Select one connecting edge for each interblock interface in path order. If a singleton block already has an incoming selected edge, choose a different outgoing one, which is possible from the two candidates. In a nonsingleton block the two selected edges are in its distinct end fibres, so cannot conflict.

Use Lemma 5.9 to construct each nonsingleton block cycle preserving its selected end edges; singleton blocks already have their fixed cycles. Splice the resulting disjoint block cycles along the selected interfaces. Each step joins two cycles into one, and every later selected edge is still present. The one-block case needs no final splice. All transitions are legal: a local step changes one path vertex, and a matching cross-edge performs its actual bottom recolouring while holding the local colouring fixed. Finally the fibres are disjoint and exhaust all proper colourings. Thus the resulting single cycle is Hamiltonian, including its closing edge. □

Theorem 5.11. *Let G be a nonempty finite simple planar graph of girth at least five and maximum degree at most three. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Disconnected graphs are allowed. The uniform Hamiltonian threshold of this class is exactly four; four-colour Hamilton-connectedness is not asserted.*

We prove a local restoration lemma using only Hamiltonicity of the smaller recolouring graph, and then supply the planar structure. The new literal words, pools and all 313 interface records appear in Appendix E. They are explicit finite identities in the proof, not references to solver verdicts. The unrestricted planar girth-five four-colour question remains open.

5.6 Listed pentagon factors

Lemma 5.12. *If every list on an induced pentagon is a subset of Q of size at least three, its list recolouring graph has a spanning factor whose components are edges, triangles, or 15-cycles.*

Proof. When all lists have size three, record their omitted colours as t. Up to the dihedral action and a common palette permutation, the boundary types and factors are listed in Table D.1 in Appendix D. The multiplicities $5, 4+1, 3+2, 3+1+1, 2+2+1, 2+1+1+1$ yield, respectively, $1, 1, 2, 2, 3, 2$ position types on a pentagon. Appendix D prints every factor: semicolons separate blocks, and an unseparated line is a single block. Successive words differ at one coordinate (cyclically for a triangle or 15-cycle), and all words are legal and distinct. For completeness the orders can be checked from

$$|F_t| = \operatorname{tr}(D_{t_0} A D_{t_1} A D_{t_2} A D_{t_3} A D_{t_4} A), \qquad A = J_4 - I_4,$$

where D_c has diagonal zero at c and one elsewhere. Expanding the trace counts exactly the legal words. The 590 printed legal, distinct words have these orders, proving coverage by explicit finite identities. If a list is all of Q, fix the other four colours. The remaining vertex has either two or three available colours, a clique fibre. These fibres directly partition the state space. □

5.7 The finite protected-edge identities

Put $Q = \{0, 1, 2, 3\}$. A boundary t has five entries in Q, with at most one entry $*$. Its local graph F_t consists of the proper C_5 colourings x satisfying $x_i \neq t_i$; $*$ excludes no colour. In the certificates $*$ is encoded by 4. The eleven ordinary types have 590 states in total. The constant type has two components, each a 15-cycle; all other ordinary types are connected. Eight have the printed Hamilton cycles, and the two remaining types are 00001 and 00112. With a full list, the eleven types, after rotation, reflection and colour permutation, are

$$0000*,\ 0001*,\ 0010*,\ 0011*,\ 0012*,\ 0101*,\ 0102*,\ 0110*,\ 0112*,\ 0120*,\ 0123*.$$

Their printed Hamilton cycles have respectively $54, 60, 78, 62, 71, 88, 85, 74, 74, 83, 82$ states, a total of 811. Thus there are 23 component types, including the two constant components.

For a Hamilton component A with specified cycle C_A, designate an ordered pool of eight edges of C_A, and let K_A be the other cycle edges. The first two pool edges will join consecutive copies of C_A; the last six will join factor blocks. For a hard component, specify cycles of $A \square P_2$ and $A \square P_3$, and a set D_A of internal edges occurring in *every* row of both seeds. We use new seeds for 00001 with $|D_A| = 19$ and the earlier seeds for 00112 with $|D_A| = 18$. Designate a seven-edge pool in D_A, and put $K_A = D_A \setminus \text{pool}$. Its first edge joins successive seed blocks and its last six join factor blocks. All pool edges are distinct.

Here is the exact normalization convention. Enumerate $s = 0, \ldots, 4$, then $\epsilon = +1, -1$, then the permutations $p \in S_4$ in lexicographic order. Take the first standard type $n_i = p(t_{s+\epsilon i})$, retaining $*$. Restoration puts the standard entry x_i at position $s + \epsilon i$ with colour $p^{-1}(x_i)$. In particular a standard boundary first normalizes by the identity.

Lemma 5.13 (Protected-edge identities)**.** *Let the source be a standard component A of F_t, and change one non-star boundary entry. Normalize the target by the preceding rule. Every target component B whose restored state set meets A has an edge in common with K_A after restoration of K_B. For an unchanged boundary use the same component and the same map.*

Proof. The ordinary and starred orbit representatives cover $4^5 + 5 \cdot 4^4 = 2304$ labelled boundaries. The displayed cyclic words give their specified component cycles and the two hard seeds: their entries are precisely the legal words, without repetition, and successive entries, including last and first, differ in one coordinate. Every listed common edge occurs in all five seed rows. Removing the listed pools therefore defines the asserted protected sets. For each standard source component, each non-star position, each different new excluded colour and each overlapping normalized target component, the appendix gives an actual edge in both protected sets. There are 313 records. Checking membership of the two five-digit endpoints in the specified cycles and outside their pools proves each record directly. Applying the inverse coordinate and colour maps gives every labelled case. This is a complete finite case proof; no finite-length gluing experiment is used to infer a general rule. □

5.8 Products retaining protection in every row

Lemma 5.14. *For arbitrary lists contained in Q, a listed C_5 recolouring graph has maximum degree at most seven.*

Proof. In a proper four-colouring, a vertex has one possible move, or two when its two neighbours have the same colour. Since $\alpha(C_5) = 2$, each colour occurs at most twice and at most two colours are repeated. A repeated pair is the neighbour pair of exactly one vertex. Hence there are at most two equal-neighbour corners and at most $5 + 2 = 7$ moves. Restricting the lists only deletes moves. □

Lemma 5.15. *For any two of the 23 component types A, B, the graph $A\square B$ has a Hamilton cycle retaining a copy of every edge of K_A in every physical B row.*

Proof. First construct $A\square P_r$ for every $r \geq 2$. When C_A exists, join consecutive copies of it by alternating the first two pool edges. At an intermediate row the two used edges are distinct. Every join cuts one edge in two disjoint circles and joins the resulting paths, giving a single circle. When A is hard, write r as a sum of twos, preceded by one three if r is odd. Use the specified seeds on these intervals and join successive intervals at their end rows using the first pool edge. Every interval has at least two rows, so its two interfaces use different physical rows. Both constructions retain K_A and the six remaining pool edges in every row.

Partition B into ordered paths with at least two vertices: use its edge/triangle factor in an ordinary nonconstant type, one 15-cycle cut into a path in a constant component, or its printed Hamilton cycle cut into a path in a starred type. Contract these paths and choose a spanning tree in the quotient, with an actual edge of B witnessing each tree edge. Every physical vertex has at least one neighbour within its factor path. By Lemma 5.14, it witnesses at most six tree edges. Start with the path-product circles and join them in parent-before-child order. The parent row has consumed at most five of the six remaining pool edges, while the fresh child row has all six. Choose a common unused edge for the two-edge switch. Every step gives one circle on the blocks processed so far, and never removes an edge of K_A. The final circle covers $A\square B$. □

Lemma 5.16. *For a fixed star position, or no star, let two boundaries be equal or differ in one non-star position. Any overlapping component pair has at least ten common vertices.*

Proof. An unchanged constant component has 15 vertices. Between a constant and nonconstant boundary, one additional colour is excluded at one coordinate of that 15-cycle. A cyclic permutation of its three colours preserves each component: an order-three action on a two-element component set is trivial. Hence each colour appears five times at each coordinate and precisely ten states remain. In every other case, both components are their entire local state sets. The intersection lists have size at least three except possibly one list of size two. Start at that position, proceed around the cycle, and finally avoid the two already coloured neighbours. This gives at least $2 \cdot 2^3 \cdot 1 = 16$ colourings. With no changed position the bound only improves. Full lists cause no difficulty. □

Consequently, if each of two boundaries changes at most one entry, normalize the target *relative to the source maps*. The two reference product cycles have at least ten common internal edges: one left protected edge from Lemma 5.13, copied at the ten or more common right states from Lemma 5.16. The two coordinate permutations and colour permutations may be independent. Actual synchronous changes are included in this stronger local classification; physical exterior vertices are never identified merely because they have equal colours.

5.9 An arbitrary bottom cycle

Lemma 5.17. *Suppose G has girth at least five and contains two vertex-disjoint 5-cycles with no edge between them. Each cycle has vertices of degree three except for at most one vertex of degree two. Let H be obtained by deleting both cycles. If $\mathcal{R}_4(H)$ is Hamiltonian, so is $\mathcal{R}_4(G)$. Neither planarity nor three-colourability of H is assumed here.*

Proof. Each degree-three cycle vertex has one exterior neighbour; a degree-two vertex gives a full list. An exterior vertex meets a given 5-cycle at most once, since two attachments would create a 3- or 4-cycle. Thus a bottom recolouring changes at most one list in each factor. The two factors may change simultaneously. All exterior neighbours remain in H, which is nonempty.

Along a bottom Hamilton cycle, every ordinary pentagon has a nonconstant boundary somewhere. Otherwise consecutive constant five-tuples, differing in at most one entry, must be equal. Its entire boundary would stay fixed along the bottom cycle. A global colour permutation changing that colour supplies another proper colouring of H which the Hamilton cycle must visit, a contradiction. These nonconstant layers need not coincide for the two factors.

Make a component graph whose vertices are all product components in all bottom layers, joining components in consecutive layers when they have a common local state. There are at most four components per layer and every component-graph degree is at most eight. This graph is connected. Indeed, on each constant run the two 15-cycle components can be labelled consistently by a binary bit: consecutive constant boundaries are equal. A nonconstant layer forgets, or resets, that bit. A starred factor resets it at every layer. Take four bit-labelled copies of the bottom cycle and allow a flip of each bit at its reset layers. This graph is connected because each bit has a reset somewhere. Contract the reset edges within each layer. The result is exactly the component graph: intersections between nonconstant components are nonempty, each constant component meets its nonconstant neighbour, and an unchanged constant component meets precisely itself.

Choose a spanning tree of the component graph. Give the root one product cycle from Lemma 5.15 and record its two restoration maps. For a child, apply the inverse parent maps to the target boundaries, normalize, then compose restoration maps. The target component is fixed by any common state. This makes the source a standard type for the protected-edge table, so the two *reference* product cycles share at least ten internal edges. The parent's component has tree degree at most eight and has so far lost at most seven internal edges. The child is fresh. Choose an unused common edge xy, delete its two copies and add the cross-layer edges at x and y. Each cross edge is

the actual bottom recolouring with the complete local colouring unchanged. The two circles become two paths joined into one circle. Induction over the tree preserves one circle, disjoint coverage and legality, and ends with all extensions of all bottom colourings. There is no additional unproved closing step. □

5.10 The planar structural reduction

Lemma 5.18. *Every connected plane graph of girth at least five with $2 \le d(v) \le 3$ and at most one degree-two vertex contains two vertex-disjoint 5-cycles with no edge between them.*

Proof. For any connected cyclic plane graph of girth at least five and maximum degree three, let p and q count vertices of degrees one and two. Euler's identity, $2e = 3n - 2p - q$, and the face-length sum give

$$f_5 = 12 - 4p - 2q + \sum_{|F|\ge 7} (|F| - 6). \tag{*}$$

Every facial walk contains a cycle and so has length at least five; bridges count twice. A length-five facial walk is a simple 5-cycle. For the graph in the lemma, $p = 0, q \le 1$, hence $f_5 \ge 12 - 2q$.

If there is a bridge ab, take the side C not containing the unique degree-two vertex, if it exists. In C, only a has degree two and all other vertices have degree three. This side is cyclic, and (*) gives at least ten 5-faces. At most two contain a. Choose a 5-cycle S avoiding a. The other bridge side has either at most two degree-two vertices, or one degree-one vertex when b was the original degree-two vertex. All other vertices have degree three. The degree sum rules out a tree and (*) gives at least eight 5-faces. Any such cycle P is disjoint from S and has no edge to it, because ab was the only edge between the sides.

Assume now there is no bridge. The subcubic graph is 2-connected: at a cutvertex every branch would require at least two incident edges to avoid a bridge, forcing degree at least four. Take a facial 5-cycle S and let B be its exterior neighbours. Girth makes them distinct; $b = |B|$ is five, or four when S contains the unique degree-two vertex z. Set $H = G - S$, embedded with the deleted region outside. Every member of B is on that outside face. Its degree drops from three to two, except that $z \in B$ would become a leaf. There is at most one leaf and no isolated vertex, so every component of H is cyclic.

In one such component, let r count its members of B, $q' \le 1$ its original degree-two vertices, and ℓ its outside facial length. Formula (*) after removing the outside-face contribution is

$$f_5^{\text{in}} = \ell + 6 - 2r - 2q' + L, \qquad L = \sum_{\substack{F \text{ internal} \\ |F|\ge 7}} (|F| - 6) \ge 0. \tag{**}$$

Even when $z \in B$, the degree deficit $4p + 2q$ remains $2r + 2q'$, so the formula still applies. Here $\ell \ge 5$. A member of B belongs to at most one internal 5-face: its degree is at most two, with an outside face angle, and a leaf belongs to none. If H is disconnected, some component has $r \le 2$ since $|B| \le 5$. By (**) it has at least $9 - 2r > r$ internal 5-faces. One avoids B, and is the required P.

Suppose H is connected. The b exterior edges partition the outside angles along S into b face portions, so at most b other faces are involved. Deleting five vertices and $5 + b$ edges decreases the face count by b; thus these b neighbours $F_1, \dots, F_b$ are distinct. Their merger has outside length

$$\ell = \sum_{i=1}^{b} |F_i| - (2b + 5), \tag{***}$$

because the five cycle edges are removed once and the b outgoing edges twice from that sum.

If $q = 0$, any S has $b = 5$, $\ell \geq 10$, $q' = 0$. Formula (**) gives at least six internal 5-faces, more than $|B|$. If z lies on a 5-face, choose that face as S. Now $b = 4$, $\ell \geq 7$, $q' = 0$, and there are at least five, again more than $|B|$.

It remains that z lies on no 5-face. There is a 5-face adjacent to two distinct non-5-faces. To see this, consider the plane dual. It is loopless because the original graph is bridgeless. All its faces are triangles except one digon corresponding to z. The distinct endpoints of that digon both represent non-5-faces. Let W be the set of all such dual vertices. If every vertex outside W had at most one distinct neighbour in W, a triangular face incident with a W–W edge would have its third vertex in W. Starting with a digon edge and going around either endpoint therefore shows that all its neighbours are in W: a triangular face propagates the property to the next neighbour, and the digon merely repeats the same neighbour. Repeating at any reached neighbour makes W closed under adjacency. Dual connectivity would force all vertices into W, contradicting $f_5 \geq 10$. This proves the assertion. Choose the resulting 5-face S. If its deletion disconnects the graph, use the case already proved; otherwise $b = 5, q' = 1$ and two distinct neighbours in (***) have length at least six. Thus $\ell \geq 12$ and (**) again gives at least six internal 5-faces. In every case one avoids B, making it disjoint and nonadjacent to S. □

5.11 Induction, checks and the remaining frontier

Proof of Theorem 5.11. Induct on the number of vertices, allowing disconnected graphs. The one-vertex case is $\mathcal{R}_4(G) = K_4$. A vertex of degree at most one is handled by Lemma 2.2. If there are two degree-two vertices, adjacent ones use Lemma 5.5. Ones at distance two use Lemma 5.10, unless their middle vertex has degree two, when an adjacent pair already applies. Ones at distance at least three use Lemma 5.6, whose spare-colour witness uses Grötzsch's theorem [6]; the new pentagon lift does not. Pairs in different components have disjoint neighbourhoods and are covered by the same separated-pair argument.

The remaining graph has minimum degree at least two and at most one degree-two vertex. Apply Lemma 5.18 inside any component. Delete its two separated pentagons. The remaining graph is nonempty and belongs to the same class, so induction gives its four-colour Hamilton cycle. Lemma 5.17 restores both pentagons. This proves the assertion at four colours. For every $k \geq 5$ use the triangle-free planar theorem proved earlier, with larger colours separately supplied there by degeneracy. The frozen 012012 colouring of C_6 gives the Hamiltonian threshold lower bound four. No monotonicity in k is presumed. □

5.12 Block-subcubic lemma

Lemma 5.19. *Let G be a nonempty finite simple planar graph of girth at least five. If every block B satisfies $\Delta(B) \leq 3$, then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Cutvertices may have arbitrarily large degree in G. The uniform Hamiltonian threshold of this class is exactly four. Four-colour Hamilton-connectedness is not asserted.*

Proof. Treat bridges as two-vertex blocks and isolated vertices as singleton blocks. The class is closed under vertex deletion: a nontrivial two-connected block of a subgraph is a two-connected subgraph of G, hence is contained in one original block, with no larger internal degrees. Planarity and girth are also preserved.

Products of Hamiltonian graphs are Hamiltonian: copy one factor cycle along a spanning path of another factor cycle, and merge successive copies using two alternating, distinct internal edges. This also handles disconnected original graphs. Choose a smallest four-colour counterexample, which may therefore be assumed connected. The singleton base is K_4, and the single-vertex lift excludes

vertices of degree at most one. If there is only one block, the previous subcubic girth-five theorem applies. Thus assume minimum degree at least two and at least one cutvertex.

Lemmas 5.5, 5.10, and 5.6 exclude adjacent degree-two vertices, a $(2,3,2)$ path, and two degree-two vertices at distance at least three. Consequently any two degree-two vertices not already giving a reduction must have a common neighbour of degree at least four in G.

The block-cut tree has at least two endblocks. No endblock is a bridge, since its noncut endpoint would be a leaf. Let B be an endblock and x its unique cutvertex of G. Every vertex of $B - x$ has its full G-degree in B, hence degree two or three; also $d_B(x) \in \{2,3\}$. Let q count the degree-two vertices of $B - x$ and put $\varepsilon = [d_B(x) = 2]$. Euler's identity in B gives

$$f_5(B) = 12 - 2(q + \varepsilon) + \sum_{|F| \geq 7} (|F| - 6).$$

Each five-face is a simple pentagon, and at most $d_B(x) \leq 3$ five-faces meet x.

If $q \leq 1$, then $f_5(B) \geq 8 > 3$, so choose a five-face avoiding x. Its vertices have degree at most three in G, with at most one degree-two vertex. If $q \geq 2$, take any two degree-two vertices of $B - x$. Their neighbours all lie in B. In the absence of the three preceding reductions, a common neighbour has G-degree at least four; the only possible such vertex in this endblock is x. Thus all q degree-two vertices are adjacent to x, and $q \leq d_B(x) \leq 3$. In either possible degree of x this gives $q + \varepsilon \leq 3$, whence $f_5(B) \geq 6 > 3$. A five-face avoiding x also avoids all those degree-two vertices, because a cycle through any one of them must use its edge to x.

Choose these pentagons in two distinct endblocks. Their vertices are noncutvertices of G and belong to their respective endblocks only, so the pentagons are disjoint and have no edge between them. The established protected-edge lift for two separated pentagons applies, including its allowance of one degree-two vertex per pentagon and simultaneous exterior list changes. The smaller graph remains in the class and has a bottom Hamilton cycle by induction. Restoration contradicts minimality.

For every $k \geq 5$ separately use the established triangle-free planar theorem. Finally C_6 belongs to the class and its periodic 012012 colouring is frozen in $\mathcal{R}_3(C_6)$, giving the lower bound four. No monotonicity in the number of colours has been assumed. □

In particular, a remaining minimum counterexample for general planar girth-five four-colour Hamiltonicity must contain a block with a vertex of degree at least four *inside that block*. High total degree accumulated only at cutvertices is insufficient. This does not assert that every minimum counterexample is two-connected.

6 Sharpness and the unresolved frontier

The lower bounds used above are frozen colourings, not claims about a particular embedding.

Proposition 6.1 (Sharpness examples)**.** *The following obstructions hold.*

(i) The icosahedral graph has a frozen proper six-colouring (see Bousquet–Perarnau [3]), so $\mathcal{R}_6$ is disconnected. Hence the planar threshold is at least seven.

(ii) The cube has a frozen proper four-colouring (identify colours with $\mathbb{Z}_2^2$ and use the three nonzero edge directions), so triangle-free and bipartite planar classes have threshold at least five.

(iii) A cycle whose length is a multiple of three has a frozen periodic three-colouring. This gives the lower bound four for girth at least six and for the four-colour girth-five subclasses containing C_6.

(iv) $\mathcal{R}_3(P_3)$ *at* $k = 3$ *has twelve states; its six degree-two states force three disjoint four-cycles, so it has no spanning cycle. Also* $\mathcal{R}_3(C_5) = C_{15} \sqcup C_{15}$. *Thus the forest and* C_5 *thresholds cannot be three.*

Proof. For (i)–(iii), every vertex sees all colours other than its own, so the displayed colouring is frozen. For (iv), the forced-edge description of $\mathcal{R}_3(P_3)$ gives three disjoint four-cycles, while the thirty proper three-colourings of C_5 have degree two and split into two components of length fifteen. □ □

The general planar girth-five problem is deliberately left at the following boundary:

$$4 \le h(\mathcal{P}_5) \le 5, \qquad 4 \le \mathrm{hc}(\mathcal{P}_5) \le 5,$$

where $\mathcal{P}_5$ is the class of planar graphs of girth at least five. The upper bounds are the five-colour Hamilton-connected theorem. The lower bounds record the four-colour boundary; no complete-state-graph counterexample has been verified in the present record.

The following implications are invalid and are not used here:

- a connected or 2-connected recolouring graph need not have a Hamilton cycle;
- a Hamilton cycle in a fixed list fibre need not extend to the complete recolouring graph;
- a spanning two-factor, a quotient cycle, or a finite local kernel does not by itself give a single Hamilton cycle;
- the four-colour result for at most seven five-cycles or for subcubic graphs does not settle all planar graphs of girth at least five.

7 Verification note

The finite data used for the endpoint-sensitive statements were replayed in fresh output directories. The local Hamilton-connected kernel for the triangle-free five-colour induction consists of 53 representatives, 225 labelled list pairs, and 3225 legal interfaces; the independent reader checked 27,600 entry cases. These numbers certify the displayed finite local templates only. The complete C_5 certificate contains 240 states, 478 prescribed-endpoint paths, and 114,242 single-coordinate steps; the two three-colour components were read back as 15-cycles. For the protected pentagon lift, the fresh readers covered 590 ordinary factor words, 465 factor cycles, 475 seeds, 1024 boundary records, and 245,760 labelled states; the protected reader covered 23 component types, 2304 labelled boundaries, and 313 interfaces, with at least ten common states per interface. The block-structure reader checked 22 structures, including a candidate whole graph with 1,495,752 states. These are bounded certificate replays, not arbitrary-size graph searches.

All arbitrary-order assertions in this paper are supplied by the analytic inductions: the degenerate theorem, the separated-degree arguments in the planar Hamilton-connected proofs, the Euler reductions for the few-five-cycle class, and the protected-edge restoration for the subcubic classes. The finite checks do not constitute external peer review, formal verification, or an extrapolation from sampled graphs. The current research record keeps the complete planar girth-five four-colour Hamiltonian and Hamilton-connected statements as unresolved.

A Summary of threshold results

Table A.1 collects the conclusions proved in the main text. A range records a proved lower and upper bound; the four-colour statement between them remains unresolved.

Table A.1: Hamiltonian and Hamilton-connected thresholds for the graph classes considered.

Graph class	h	hc	additional conclusion
All planar graphs	7	7	$k \geq 8$: vertex-pancyclic
Triangle-free planar graphs	5	5	$k \geq 6$: vertex-pancyclic
Bipartite planar graphs	5	5	the cube gives the four-colour lower bound
Planar graphs with $g(G) \geq 6$	4	4	$k \geq 5$: vertex-pancyclic
Nonempty forests	4	4	$k \geq 4$: vertex-pancyclic
The single graph C_5	4	4	$\mathcal{R}_3(C_5)$ is two 15-cycles
Planar $g \geq 5$, at most seven distinct 5-cycles	4	$4 \leq hc \leq 5$	four-colour Hamiltonian
Planar $g \geq 5$, $\Delta \leq 3$	4	$4 \leq hc \leq 5$	four-colour Hamiltonian
General planar $g \geq 5$	$4 \leq h \leq 5$	$4 \leq hc \leq 5$	the four-colour problem is open

B Adjacent-pair endpoint certificates

These certificates use the graph $F(A,B)$ and the type $(|A|,|B|,|A\cap B|)$ defined before Lemma 4.3. The two-digit word ab denotes (a,b). Each displayed word lists all states once, with consecutive states differing in one coordinate; the 333 word is cyclic. The counts in Table B.1 include all colour permutations preserving the listed pair of lists, coordinate exchange when it preserves the type, and path reversal. The proof of Lemma 4.3 gives the negative witnesses for the excluded endpoint pairs.

Table B.1: Adjacent-pair endpoint certificates and their symmetry coverage.

Type	$\lvert V(F)\rvert$	Words	Feasible unordered pairs
331	8	6	28
332	7	7	19
333	6	1	6
342	10	16	45
343	9	8	36
443	13	11	78
444	12	4	66

The 53 endpoint words. For 331, use $A=\{0,1,2\}$ and $B=\{0,3,4\}$:

```
03 13 23 24 20 10 14 04
03 23 20 24 04 14 13 10
03 23 20 24 04 14 10 13
03 23 13 10 20 24 04 14
13 10 20 24 23 03 04 14
13 14 10 20 23 03 04 24
```

For 332, use $A=\{0,1,2\}$ and $B=\{0,1,3\}$:

```
01 21 23 20 10 13 03
01 21 20 23 03 13 10
01 03 23 21 20 10 13
01 21 20 10 13 03 23
03 01 21 23 20 10 13
03 01 21 23 13 10 20
03 01 21 20 10 13 23
```

For 333, use $A=B=\{0,1,2\}$:

```
01 21 20 10 12 02
```

For 342, use $A=\{0,1,2\}$ and $B=\{0,1,3,4\}$:

```
01 21 24 20 23 13 10 14 04 03
01 21 23 20 24 14 04 03 13 10
01 21 24 23 20 10 14 04 03 13
01 21 23 24 14 04 03 13 10 20
01 04 14 24 23 03 13 10 20 21
```

```
01 21 24 20 10 13 14 04 03 23
03 23 24 14 13 10 20 21 01 04
03 23 24 20 21 01 04 14 10 13
03 13 10 20 24 23 21 01 04 14
03 23 24 21 01 04 14 13 10 20
03 13 23 24 20 10 14 04 01 21
03 01 21 20 24 04 14 10 13 23
03 23 13 14 10 20 21 01 04 24
20 24 23 13 10 14 04 03 01 21
20 10 13 14 04 24 21 01 03 23
23 21 20 10 14 13 03 01 04 24
```

For 343, use $A = \{0, 1, 2\}$ and $B = \{0, 1, 2, 3\}$:

```
01 21 23 20 10 12 13 03 02
01 21 20 23 13 10 12 02 03
01 21 20 23 13 03 02 12 10
01 21 23 20 10 13 03 02 12
01 21 23 20 10 12 02 03 13
01 02 03 23 13 12 10 20 21
01 21 20 10 13 12 02 03 23
03 23 20 21 01 02 12 10 13
```

For 443, use $A = \{0, 1, 2, 3\}$ and $B = \{0, 1, 2, 4\}$:

```
01 31 34 24 21 20 30 32 12 10 14 04 02
01 21 31 34 32 30 20 24 14 10 12 02 04
01 21 31 34 32 30 20 24 14 04 02 12 10
01 31 32 30 34 24 21 20 10 14 04 02 12
01 31 32 30 34 24 21 20 10 12 02 04 14
01 21 20 30 32 31 34 14 10 12 02 04 24
01 21 31 32 30 20 24 14 10 12 02 04 34
04 24 34 32 31 30 20 21 01 02 12 10 14
04 34 14 24 20 21 01 31 32 02 12 10 30
04 34 32 30 20 21 24 14 10 12 02 01 31
04 24 20 30 32 31 21 01 02 12 10 14 34
```

For 444, use $A = B = \{0, 1, 2, 3\}$:

```
01 31 21 23 20 30 32 12 10 13 03 02
01 21 31 32 30 20 23 13 03 02 12 10
01 21 31 32 30 10 20 23 13 03 02 12
01 21 31 32 30 20 10 13 12 02 03 23
```

C Certificates for the $(2,3,2)$-path restoration

The fibres B_t, allowed colour images and path reversals, and special edge sets P_t are defined in the main text. The following literal cycles and interface witnesses supply the finite input to Lemma 5.8.

C.1 The three normalized cycles

In the cyclic words below the coordinates are uvw. The rows for $001, 010, 012$ specify D_E, D_M, D_A, respectively.

$$\begin{array}{l|l}
001 & 120\ 123\ 323\ 313\ 213\ 210\ 230\ 130\ 132\ 232\ 212\ 312\ 310\ 320 \\
010 & 101\ 102\ 103\ 203\ 303\ 323\ 123\ 121\ 321 \\
 & 301\ 302\ 202\ 201\ 231\ 232\ 132\ 131 \\
012 & 101\ 121\ 120\ 320\ 321\ 323\ 123\ 103 \\
 & 203\ 303\ 301\ 201\ 231\ 230\ 130\ 131
\end{array} \tag{C.1}$$

C.2 Safe common-edge witnesses

Number source edges in (C.1) cyclically from one. Table C.1 supplies two safe common source-edge indices in every case. Target columns are defined as follows.

For an end-change target, write a for its repeated colour, b for its singleton, and $h_0 < h_1$ for the unused colours. There is one allowed edge set: apply $p = (a, b, h_0, h_1)$ to D_E, reversing the path if the singleton is at the left end. Swapping the two unused colours preserves this edge set. For a middle-change target (a, b, a) there are four images of D_M: column $\epsilon\eta$ uses $p = (a, b, h_\eta, h_{1-\eta})$ and reverses the path exactly when $\epsilon = 1$. For an all-distinct target (a, b, c) with unused colour d, columns 1 and 2 are respectively $p = (a, b, c, d)$ applied to D_A, and $p = (c, b, a, d)$ followed by path reversal. These exhaust the images in each case.

Table C.1: Safe common source-edge indices for adjacent good boundaries.

t	s	1 / 00	2 / 01	10	11
001	002	1, 3	–	–	–
001	003	6, 8	–	–	–
001	011	1, 7	–	–	–
001	021	4, 7	4, 5	–	–
001	031	1, 5	1, 3	–	–
001	101	3, 4	4, 5	3, 12	5, 10
001	201	8, 12	1, 8	–	–
001	301	1, 5	1, 5	–	–
010	012	3, 4	3, 6	–	–
010	013	9, 11	8, 10	–	–
010	020	1, 3	2, 14	3, 10	2, 4
010	030	5, 7	2, 6	1, 3	1, 2
010	210	1, 10	2, 6	–	–
010	310	3, 7	2, 13	–	–
012	013	1, 2	3, 4	–	–
012	032	3, 8	2, 4	–	–
012	312	8, 12	1, 7	–	–

D Literal pentagon factors and common-port seeds

Five digits record the five colours in cyclic vertex order. In six-digit words the first digit is the path layer. Semicolons separate factor blocks; without a semicolon the entire factor line is one block. All Hamilton words are cyclic. A port is a pair of five-digit words. These literal identities are part of the proof, not solver verdicts.

Table D.1: Spanning factors for the eleven normalized pentagon boundary types.

t	$\lvert F_t\rvert$	Factor	t	$\lvert F_t\rvert$	Factor
00000	30	$2C_{15}$	00001	44	$22K_2$
00011	42	$21K_2$	00012	47	$22K_2 + K_3$
00101	64	$32K_2$	00102	63	$30K_2 + K_3$
00112	51	$24K_2 + K_3$	00121	58	$29K_2$
00123	57	$27K_2 + K_3$	01012	68	$34K_2$
01023	66	$33K_2$			

Factor 00000 (30 words).

```
12123 12323 12313 12312 32312 31312 31212 31232 31231 21231 23231 23131 23121 23123 13123
12132 13132 13232 13212 13213 23213 21213 21313 21323 21321 31321 32321 32121 32131 32132
```

Factor 00001 (44 words).

```
12120 32120 ; 12123 12323 ; 12130 32130 ; 12132 32132 ; 12310 32310
12312 12313 ; 12320 32320 ; 13120 23120 ; 13123 23123 ; 13130 23130
13132 13232 ; 13210 23210 ; 13212 13213 ; 13230 23230 ; 21210 31210
21213 23213 ; 21230 31230 ; 21310 31310 ; 21313 21323 ; 21320 31320
31212 31232 ; 31312 32312
```

Factor 00011 (42 words).

```
12102 32102 ; 12103 12123 ; 12120 32120 ; 12130 32130 ; 12132 32132
12302 12303 ; 12320 12323 ; 13102 13202 ; 13103 23103 ; 13120 23120
13123 23123 ; 13130 23130 ; 13132 13232 ; 13203 23203 ; 13230 23230
21203 21303 ; 21230 31230 ; 21320 21323 ; 31202 31232 ; 31302 32302
31320 32320
```

Factor 00012 (47 words).

```
12103 12303 ; 12120 12320 ; 12123 12323 ; 12130 32130 ; 13103 13203
13120 13130 ; 13123 23123 ; 13230 23230 ; 21201 31201 ; 21203 23203
21230 31230 ; 21231 31231 ; 21301 21303 ; 23101 23103 ; 23120 23121
23130 23131 ; 23201 23231 ; 31301 32301 ; 31320 32320 ; 31321 32321
32101 32131 ; 32120 32121 ; 21320 21321 21323
```

Factor 00101 (64 words).

```
12010 32010 ; 12012 32012 ; 12013 12313 ; 12020 32020 ; 12023 12323
12030 32030 ; 12032 32032 ; 12310 32310 ; 12312 32312 ; 12320 32320
13010 23010 ; 13012 13212 ; 13013 23013 ; 13020 23020 ; 13023 23023
```

13030 23030 ; 13032 13232 ; 13210 23210 ; 13213 23213 ; 13230 23230
21010 31010 ; 21013 21313 ; 21020 31020 ; 21023 21323 ; 21030 31030
21210 21213 ; 21230 31230 ; 21310 31310 ; 21320 31320 ; 31012 31312
31032 31232 ; 31210 31212

Factor 00102 (63 words).

12013 12023 ; 12310 12313 ; 12320 12323 ; 13010 23010 ; 13013 23013
13020 23020 ; 13023 23023 ; 13030 13230 ; 13210 13213 ; 21010 21013
21020 21023 ; 21021 23021 ; 21030 23030 ; 21031 31031 ; 21210 31210
21213 21313 ; 21230 23230 ; 21231 31231 ; 21310 21320 ; 21321 21323
23031 23231 ; 23210 23213 ; 31010 32010 ; 31020 31320 ; 31021 32021
31030 31230 ; 31310 32310 ; 31321 32321 ; 32020 32320 ; 32030 32031
12010 12020 12030

Factor 00112 (51 words).

12020 12023 ; 12030 32030 ; 12303 12323 ; 12320 32320 ; 13020 13023
13030 13230 ; 13203 23203 ; 21030 31030 ; 21031 23031 ; 21201 23201
21203 21303 ; 21230 31230 ; 21231 31231 ; 21301 21321 ; 21320 21323
23020 23030 ; 23021 23023 ; 23230 23231 ; 31020 32020 ; 31021 32021
31031 32031 ; 31201 31301 ; 31320 31321 ; 32301 32321 ; 21020 21021 21023

Factor 00121 (58 words).

12010 32010 ; 12012 12013 ; 12030 32030 ; 12032 32032 ; 12302 32302
12303 12313 ; 12310 32310 ; 12312 32312 ; 13010 23010 ; 13012 13212
13013 13213 ; 13030 23030 ; 13032 13232 ; 13202 13203 ; 13210 23210
13230 23230 ; 21010 31010 ; 21013 23013 ; 21030 31030 ; 21203 21303
21210 21213 ; 21230 31230 ; 21310 21313 ; 23203 23213 ; 31012 32012
31032 31232 ; 31202 31302 ; 31210 31212 ; 31310 31312

Factor 00123 (57 words).

12010 12310 ; 12012 32012 ; 12030 13030 ; 12032 32032 ; 12302 12312
13010 23010 ; 13012 13032 ; 13210 13230 ; 21010 31010 ; 21030 31030
21031 23031 ; 21201 31201 ; 21210 23210 ; 21230 31230 ; 21231 31231
21301 31301 ; 21310 31310 ; 23030 23230 ; 23201 23231 ; 31012 31312
31031 32031 ; 31032 31232 ; 31202 31302 ; 31210 31212 ; 32010 32030
32301 32302 ; 32310 32312 ; 13202 13212 13232

Factor 01012 (68 words).

10103 20103 ; 10120 30120 ; 10123 20123 ; 10130 30130 ; 10203 20203
10230 30230 ; 10303 20303 ; 10320 30320 ; 10323 20323 ; 12103 12303
12120 32120 ; 12123 12323 ; 12130 32130 ; 12320 32320 ; 13103 23103
13120 23120 ; 13123 23123 ; 13130 23130 ; 13203 23203 ; 13230 23230
20101 20131 ; 20120 20320 ; 20121 23121 ; 20130 20230 ; 20201 30201
20231 30231 ; 20301 20321 ; 23101 23201 ; 23131 23231 ; 30101 32101
30121 32121 ; 30131 32131 ; 30301 32301 ; 30321 32321

Factor 01023 (66 words).

10102 30102 ; 10130 30130 ; 10132 30132 ; 10202 30202 ; 10210 30210
10212 30212 ; 10230 30230 ; 10232 30232 ; 10302 30302 ; 10310 30310
10312 30312 ; 12102 32102 ; 12130 32130 ; 12132 13132 ; 12302 32302
12310 32310 ; 12312 32312 ; 13102 13202 ; 13130 23130 ; 13210 23210
13212 13232 ; 13230 23230 ; 20101 20301 ; 20130 20230 ; 20131 30131
20201 30201 ; 20210 20310 ; 20231 30231 ; 23101 23201 ; 23131 23231
30101 32101 ; 30301 32301 ; 32131 32132

Cycle 00011 (42 words).

12102 13102 13103 23103 23123 13123 13120 23120 23130 13130
13230 23230 21230 31230 31232 31202 31302 32302 12302 12303
12103 12123 12323 12320 32320 31320 21320 21323 21303 21203
23203 13203 13202 13232 13132 12132 12130 12120 32120 32130
32132 32102

Cycle 00012 (47 words).

12103 12123 13123 13120 12120 32120 32130 12130 13130 13230
23230 21230 31230 31231 21231 23231 23201 23101 23131 23130
23120 23121 23123 23103 13103 13203 23203 21203 21201 31201
31301 32301 32101 32131 32121 32321 31321 21321 21301 21303
21323 21320 31320 32320 12320 12323 12303

Cycle 00101 (64 words).

12010 12020 32020 31020 21020 23020 13020 13010 23010 23210
13210 13212 13232 13032 13012 12012 32012 31012 31312 32312
12312 12310 32310 31310 21310 21320 31320 32320 12320 12323
12313 12013 12023 13023 23023 21023 21323 21313 21013 23013
13013 13213 23213 21213 21210 31210 31212 31232 31032 32032
12032 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21010 31010 32010

Cycle 00102 (63 words).

12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
31321 21321 21320 31320 32320 12320 12310 32310 31310 21310
21313 21323 21023 23023 13023 13020 13010 23010 21010 21020
23020 23021 21021 31021 31020 32020 12020 12023 12323 12313
12013 13013 23013 21013 21213 23213 13213 13210 23210 21210
31210 31010 32010

Cycle 00121 (58 words).

12010 13010 23010 23210 13210 13230 13030 12030 32030 31030
31032 32032 12032 13032 13232 13202 13212 13012 12012 32012
31012 31312 32312 12312 12310 32310 31310 21310 21210 21213
23213 13213 13203 23203 21203 21303 21313 21013 23013 13013
12013 12313 12303 12302 32302 31302 31202 31232 31212 31210
31230 21230 23230 23030 21030 21010 31010 32010

Cycle 00123 (57 words).

12010 12310 32310 31310 21310 21010 23010 13010 13210 13212
13202 13232 13032 13012 12012 32012 31012 31212 31232 31032
32032 12032 12030 32030 32031 31031 21031 21030 31030 31230
21230 21231 31231 31201 31202 31302 31312 32312 12312 12302
32302 32301 31301 21301 21201 23201 23231 23031 23030 13030
13230 23230 23210 21210 31210 31010 32010

Cycle 01012 (68 words).

10103 10203 13203 23203 20203 20303 10303 12303 12103 12123
10123 13123 23123 20123 20323 10323 12323 12320 32320 30320
10320 20320 20120 23120 13120 10120 30120 32120 12120 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20321 30321 32321 32121 30121 20121 23121 23101
23201 20201 30201 30101 20101 20103 23103 13103

Cycle 01023 (66 words).

10102 10302 30302 32302 12302 12312 10312 30312 32312 32310
12310 10310 30310 20310 20210 30210 10210 10212 30212 30202
10202 13202 13102 13132 13232 13212 13210 23210 23230 23130
23131 23231 23201 23101 20101 30101 30201 20201 20301 30301
32301 32101 32131 30131 30231 20231 20131 20130 20230 30230
10230 13230 13130 12130 32130 30130 10130 10132 10232 30232
30132 32132 12132 12102 32102 30102

C2 for 00001 (88 words).

012120 012130 012132 032132 032130 032120 132120 132130 132132
112132 112130 112120 112123 012123 012323 012320 112320 112323
112313 112312 012312 012313 012310 112310 132310 032310 031310
021310 121310 131310 131312 132312 032312 031312 031212 031210
031230 031232 131232 131212 131210 131230 121230 121210 121213
121313 121323 121320 131320 132320 032320 031320 021320 021323
021313 021213 021210 021230 023230 123230 113230 113210 123210
123213 113213 113212 113232 113132 013132 013232 013212 013213
023213 023210 013210 013230 013130 023130 023120 023123 123123
123120 123130 113130 113120 113123 013123 013120

C3 for 00001 (132 words).

012120 012130 012132 032132 032130 032120 032320 031320 021320
021323 021313 021213 023213 023210 013210 013230 023230 021230
121230 121210 121213 121313 121323 121320 131320 132320 132120
132130 132132 112132 112130 112120 112320 112323 112313 112310
132310 132312 112312 212312 212313 212310 212320 212323 212123
112123 012123 012323 012320 012310 012313 012312 032312 032310
031310 131310 231310 232310 232312 231312 231212 231210 231230
231232 131232 031232 031230 131230 131210 131212 131312 031312
031212 031210 021210 021310 121310 221310 221210 221230 223230
123230 113230 213230 213130 223130 223120 223123 123123 123120

123130 113130 113120 113123 213123 213120 212120 212130 212132
232132 232130 232120 232320 231320 221320 221323 221313 221213
223213 223210 213210 213213 213212 213232 213132 113132 113232
113212 113210 123210 123213 113213 013213 013212 013232 013132
013130 023130 023120 023123 013123 013120

Common ports 00001 (20 edges).

12120-12130 12130-12132 12132-32132 12320-12323 13120-13123
13130-23130 13132-13232 13210-23210 13212-13232 21213-21313
21313-21323 21320-21323 21320-31320 23120-23123 23120-23130
23210-23213 31210-31212 31320-32320 32120-32130 32130-32132

C2 for 00112 (102 words).

012020 012030 032030 032031 032021 031021 021021 021031 031031
031231 031201 031301 032301 032321 032320 012320 012323 012303
112303 112323 112320 132320 132321 132301 131301 131201 131231
131230 031230 031030 021030 021230 121230 121030 131030 131020
121020 123020 123023 123021 123031 123030 123230 113230 113030
113020 113023 112023 012023 013023 013020 013030 013230 023230
023030 023031 023231 123231 121231 021231 021201 021301 021321
031321 131321 131021 131031 121031 121021 121023 121323 121303
021303 021203 121203 123203 113203 013203 023203 023201 123201
121201 121301 121321 121320 131320 031320 021320 021323 021023
023023 023021 023020 021020 031020 032020 132020 132021 132031
132030 112030 112020

C3 for 00112 (153 words).

012020 012023 013023 013020 013030 013230 023230 021230 021030
031030 032030 032031 032021 132021 132031 132030 131030 121030
121230 123230 113230 113030 113020 113023 112023 112323 112320
132320 132321 132301 032301 032321 032320 012320 012323 012303
112303 212303 212323 212320 232320 232020 231020 221020 223020
223023 221023 221323 221320 231320 131320 031320 021320 121320
121323 121321 121301 221301 221321 231321 231021 231031 231231
231201 231301 232301 232321 232021 232031 232030 212030 212020
212023 213023 213020 213030 213230 223230 223030 223031 221031
221021 223021 123021 123023 023023 023021 023031 123031 123231
023231 023201 123201 123203 223203 213203 113203 013203 023203
021203 121203 121201 021201 031201 031301 021301 021321 031321
131321 131301 131201 131231 121231 021231 031231 031230 131230
231230 231030 221030 221230 221231 223231 223201 221201 221203
221303 121303 021303 021323 021023 121023 121021 121031 021031
021021 031021 031031 131031 131021 131020 121020 123020 123030
023030 023020 021020 031020 032020 132020 112020 112030 012030

Common ports 00112 (18 edges).

12020-12030 12023-13023 12320-12323 12320-32320 13020-13023
13020-13030 13030-13230 13230-23230 21020-23020 21020-31020
21021-21031 21030-21230 21030-31030 21301-21321 31201-31301
32021-32031 32030-32031 32301-32321

E Literal protected-pentagon certificates

Boundary digit 4 denotes the full list. Colour words use only 0,1,2,3. In six-digit cycle words the first digit is the path layer. Pool order matters: the first two edges (one for the hard types) join path layers or seed blocks; the last six join the factor tree. The protected set is the specified cycle edge set, or COMMON set for a hard type, minus POOL. The earlier appendix supplies the ordinary cycles, the two constant-type 15-cycles, and the 00112 seeds. An interface record gives source type:component, position, new colour, normalized target type:component, and the actual common edge. Positions start at zero. Two records per line are separated by semicolons.

PC2 for 00001 (88 words).

```
012120 012130 112130 112120 132120 132130 132132 112132 113132 013132
012132 032132 032130 032120 032320 032310 012310 012313 012323 012320
112320 132320 131320 031320 021320 121320 121310 021310 021313 021323
121323 121313 121213 121210 131210 131212 131232 131230 121230 123230
113230 113232 113212 113210 123210 123213 113213 013213 023213 021213
021210 023210 013210 013212 013232 013230 023230 021230 031230 031232
031212 031210 031310 031312 032312 012312 112312 132312 131312 131310
132310 112310 112313 112323 112123 012123 013123 023123 023120 023130
013130 113130 123130 123120 123123 113123 113120 013120
```

PC3 for 00001 (132 words).

```
012120 013120 023120 023130 013130 012130 112130 113130 113132 112132
132132 132130 132120 232120 232130 232132 212132 213132 213232 213230
223230 221230 231230 231232 231212 231210 231310 231312 232312 232310
212310 212312 212313 212323 212123 112123 112120 113120 213120 212120
212130 213130 223130 123130 123120 223120 223123 213123 113123 123123
023123 013123 012123 012323 012313 012312 112312 132312 032312 032310
012310 012320 032320 132320 132310 112310 112313 112323 112320 212320
232320 231320 221320 221323 221313 221310 221210 221213 223213 123213
023213 021213 021210 023210 013210 013212 013213 113213 213213 213212
213210 223210 123210 113210 113212 113232 113230 123230 121230 131230
131232 131212 131210 121210 121213 121313 121323 021323 021313 021310
121310 121320 021320 031320 131320 131310 131312 031312 031310 031210
031212 031232 031230 021230 023230 013230 013232 013132 012132 032132
032130 032120
```

Balanced common edges for 00001.

```
12132-13132 12132-32132 12310-32310 12313-12323 13123-23123
13210-13212 13210-23210 13230-13232 13230-23230 21210-21213
21230-23230 21230-31230 21313-21323 31210-31212 31212-31232
31230-31232 31310-31312 32120-32130 32130-32132
```

Cycle for 00004 (54 words).

```
12120 12130 12132 13132 13130 23130 23230 23210 13210 13230
13232 13212 13213 23213 21213 21210 31210 31212 31232 31231
31230 21230 21231 23231 23131 23121 23123 23120 13120 13123
12123 12323 12313 12312 12310 32310 32312 31312 31310 21310
21313 21323 21321 21320 31320 31321 32321 32121 32131 32132
32130 32120 32320 12320
```

Cycle for 00014 (60 words).

12102 13102 13132 13232 13202 13203 13103 12103 12123 13123
13120 12120 32120 32320 12320 12323 12303 12302 32302 31302
31202 31232 31230 21230 23230 13230 13130 23130 23120 23121
23123 23103 23203 23201 23101 23131 23231 21231 31231 31201
21201 21203 21303 21323 21320 31320 31321 21321 21301 31301
32301 32321 32121 32101 32131 32130 12130 12132 32132 32102

Cycle for 00104 (78 words).

12010 12030 32030 31030 31230 21230 23230 13230 13030 23030
21030 21031 23031 23231 21231 31231 31031 32031 32021 32321
32320 12320 12323 12313 12013 12023 13023 23023 21023 21020
31020 32020 12020 13020 23020 23021 21021 31021 31321 31320
21320 21321 21323 21313 21310 31310 32310 12310 12312 32312
31312 31012 32012 12012 13012 13013 23013 21013 21213 23213
13213 13212 13232 13032 12032 32032 31032 31232 31212 31210
21210 23210 13210 13010 23010 21010 31010 32010

Cycle for 00114 (62 words).

12020 12320 32320 31320 21320 21323 21023 23023 23021 23031
21031 21021 21020 21030 21230 31230 31030 32030 12030 13030
13230 23230 23030 23020 13020 13023 12023 12323 12303 12302
32302 31302 31202 31232 31032 32032 12032 13032 13232 13202
13203 23203 21203 21303 21301 21321 31321 32321 32301 31301
31201 21201 23201 23231 21231 31231 31031 32031 32021 31021
31020 32020

Cycle for 00124 (71 words).

12010 12030 32030 32031 31031 31231 21231 21031 21030 31030
31230 21230 23230 13230 13030 23030 23031 23231 23201 23203
21203 21303 21301 21201 31201 31202 31302 31301 32301 32302
12302 12303 12313 12013 13013 13010 13210 13212 13012 12012
32012 31012 31312 32312 12312 12310 32310 31310 31210 31212
31232 31032 32032 12032 13032 13232 13202 13203 13213 23213
21213 21313 21013 23013 23010 23210 21210 21310 21010 31010
32010

Cycle for 01014 (88 words).

10102 10132 13132 13232 10232 30232 30132 32132 12132 12130
32130 30130 10130 10230 30230 20230 23230 13230 13130 23130
20130 20131 23131 23231 20231 30231 30131 32131 32101 32301
30301 20301 20101 30101 30201 20201 23201 23101 23121 20121
30121 32121 32321 30321 20321 20323 10323 12323 12320 32320
30320 10320 20320 20120 23120 13120 12120 32120 30120 10120
10123 12123 13123 23123 20123 20103 20203 20303 10303 10302
30302 32302 12302 12303 12103 10103 10203 13203 23203 23103
13103 13102 13202 10202 30202 30102 32102 12102

Cycle for 01024 (85 words).

10102 10202 30202 30212 10212 13212 13210 23210 20210 30210
10210 10213 10203 20203 23203 13203 13213 23213 20213 20313
10313 12313 12312 12310 32310 30310 20310 10310 10312 30312

32312 32302 30302 10302 10303 20303 20103 23103 13103 10103
12103 12303 12302 12102 32102 30102 30101 30201 20201 23201
23101 20101 20301 30301 32301 32101 32131 30131 30231 20231
23231 23131 20131 20130 23130 13130 13230 23230 20230 30230
10230 10130 30130 32130 12130 12132 32132 30132 30232 10232
10132 13132 13232 13202 13102

Cycle for 01104 (74 words).

10210 10310 20310 30310 32310 12310 12312 10312 30312 32312
32012 32032 32031 32021 32321 30321 20321 20320 10320 30320
32320 12320 12020 12010 32010 32020 32030 12030 12032 13032
13232 13212 13012 12012 12013 12313 10313 20313 20323 10323
12323 12023 13023 23023 23013 13013 13213 23213 20213 10213
10212 10232 10230 13230 23230 23030 13030 13020 23020 23021
23031 23231 20231 30231 30232 30212 30210 30230 20230 20210
23210 23010 13010 13210

Cycle for 01124 (74 words).

10202 10203 20203 23203 23201 20201 20301 20303 10303 12303
12302 32302 32301 30301 30201 30231 20231 20230 30230 10230
13230 23230 23231 23031 23030 13030 12030 32030 32031 32032
32012 32010 12010 12012 12032 13032 13232 10232 30232 30202
30212 10212 10213 10210 30210 20210 20310 30310 10310 12310
32310 32312 12312 12313 12013 13013 23013 23010 23210 13210
13010 13012 13212 13202 13203 13213 23213 20213 20313 10313
10312 30312 30302 10302

Cycle for 01204 (83 words).

10120 20120 23120 13120 13130 23130 20130 10130 30130 32130
12130 12132 32132 30132 10132 13132 13032 12032 32032 32012
12012 13012 13010 12010 32010 32020 12020 13020 23020 23010
23030 13030 12030 32030 32031 32131 30131 20131 23131 23031
23021 23121 20121 30121 32121 32021 32321 30321 20321 20320
10320 30320 32320 12320 12323 12023 13023 23023 23013 13013
12013 12313 12310 32310 32312 12312 10312 30312 30310 10310
20310 20313 10313 10323 20323 20123 23123 13123 10123 12123
12120 32120 30120

Cycle for 01234 (82 words).

10102 10302 30302 32302 12302 12303 10303 10323 20323 20303
20103 10103 12103 12123 13123 23123 20123 10123 10120 30120
32120 12120 13120 23120 20120 20320 10320 30320 32320 12320
12310 32310 30310 10310 20310 20313 10313 10312 30312 32312
12312 12313 12323 12023 13023 23023 23013 23010 13010 12010
12013 13013 13012 12012 32012 32010 32020 12020 13020 23020
23021 23121 20121 30121 32121 32021 32321 30321 20321 20301
30301 32301 32101 30101 20101 23101 23103 13103 13102 12102
32102 30102

Pool for 00000:0.

12123-12323 12123-13123 12312-12313 12313-12323
13123-23123 21231-31231 23121-23131 23131-23231

Pool for 00000:1.

13132-13232 13212-13213 21213-21313 21213-23213
21313-21323 21321-31321 32121-32131 32121-32321

Pool for 00001:0.

13210-13212 21210-21213 21230-31230 31210-31212
31230-31232 31310-31312 32120-32130

Pool for 00011:0.

12102-13102 13102-13103 13123-23123 13130-13230
13202-13203 21203-21303 31202-31302 32130-32132

Pool for 00012:0.

13130-13230 21301-21321 23120-23130 23121-23123
31201-31301 32101-32301 32121-32131 32121-32321

Pool for 00101:0.

12010-12020 12012-13012 12013-12023 12320-12323
13013-23013 13210-13212 21010-31010 23010-23210

Pool for 00102:0.

12013-13013 12313-12323 13013-23013 21013-21213
21023-21323 21031-23031 23020-23021 31031-31231

Pool for 00112:0.

12020-12030 12320-12323 13020-13030 21020-23020
21021-21031 21030-31030 32021-32031

Pool for 00121:0.

12010-13010 21010-21030 21010-31010 21210-21213
23010-23210 23030-23230 31012-31312 31030-32030

Pool for 00123:0.

13030-23030 13230-23230 21010-21310 21030-31030
21230-21231 31010-31210 31012-32012 31030-31230

Pool for 01012:0.

10103-13103 10123-13123 10130-30130 13130-13230
20101-30101 20230-30230 20301-20321 30131-30231

Pool for 01023:0.

12130-13130 20101-30101 20201-20301 20210-30210
20231-30231 23130-23131 30102-32102 30202-30212

Pool for 00004:0.

12312-12313 13130-13132 13212-13213 21210-21213
21321-21323 23121-23123 31231-31232 31310-31312

Pool for 00014:0.

12102-32102 12103-13103 12120-32120 12130-12132
13130-23130 21201-21203 23121-23123 32101-32121

Pool for 00104:0.

13012-13013 13210-23210 13212-13213 21020-21023
21320-21321 21321-21323 31320-31321 32320-32321

Pool for 00114:0.

12030-13030 21020-21021 21301-21321 21320-21323
23021-23023 31020-31021 31020-32020 32021-32031

Pool for 00124:0.

12310-32310 13210-13212 21030-31030 21201-31201
21213-21313 23010-23210 31010-32010 31301-31302

Pool for 01014:0.

10120-10123 10202-13202 10302-10303 12120-13120
12123-13123 13102-13103 20130-20131 20321-20323

Pool for 01024:0.

10103-13103 10132-13132 10302-10303 10312-30312
12130-12132 12312-12313 20103-20303 20213-20313

Pool for 01104:0.

10210-10310 12010-12020 12012-12013 12310-12312
13013-13213 20210-23210 23030-23230 32012-32312

Pool for 01124:0.

10202-10203 10210-10213 10230-13230 10310-30310
13010-13210 20210-20310 23230-23231 32310-32312

Pool for 01204:0.

10123-13123 12010-32010 12120-12123 12310-12313
12320-12323 23010-23020 30310-30312 32030-32031

Pool for 01234:0.

10102-30102 12102-32102 12123-13123 12302-32302
12310-12320 13012-13013 13102-13103 20101-30101

All 313 single-pentagon interfaces.

```
00000:0 0 1 00001:0 23121-23123 ; 00000:0 0 2 00001:0 31231-31232
00000:0 0 3 00001:0 21231-23231 ; 00000:0 1 1 00001:0 12312-32312
00000:0 1 2 00001:0 31212-31312 ; 00000:0 1 3 00001:0 31231-31232
00000:0 2 1 00001:0 21231-23231 ; 00000:0 2 2 00001:0 31312-32312
00000:0 2 3 00001:0 31212-31232 ; 00000:0 3 1 00001:0 31231-31232
00000:0 3 2 00001:0 31231-31232 ; 00000:0 3 3 00001:0 23121-23123
00000:0 4 1 00001:0 31212-31232 ; 00000:0 4 2 00001:0 21231-23231
00000:0 4 3 00001:0 12312-32312 ; 00000:1 0 1 00001:0 32131-32132
00000:1 0 2 00001:0 31321-32321 ; 00000:1 0 3 00001:0 12132-13132
00000:1 1 1 00001:0 13213-23213 ; 00000:1 1 2 00001:0 21321-21323
00000:1 1 3 00001:0 12132-32132 ; 00000:1 2 1 00001:0 13212-13232
00000:1 2 2 00001:0 12132-13132 ; 00000:1 2 3 00001:0 13212-13232
00000:1 3 1 00001:0 21321-21323 ; 00000:1 3 2 00001:0 32131-32132
00000:1 3 3 00001:0 21321-21323 ; 00000:1 4 1 00001:0 12132-13132
00000:1 4 2 00001:0 13213-23213 ; 00000:1 4 3 00001:0 13212-13232
00001:0 0 1 00011:0 21230-23230 ; 00001:0 0 2 00012:0 12132-13132
00001:0 0 3 00012:0 12132-13132 ; 00001:0 1 1 00101:0 12132-32132
00001:0 1 2 00102:0 21313-21323 ; 00001:0 1 3 00102:0 12313-12323
00001:0 2 1 00101:0 12310-32310 ; 00001:0 2 2 00102:0 12132-13132
00001:0 2 3 00102:0 12132-13132 ; 00001:0 3 1 00011:0 12132-13132
00001:0 3 2 00012:0 12132-13132 ; 00001:0 3 3 00012:0 13123-23123
00001:0 4 0 00000:0 31212-31232 ; 00001:0 4 0 00000:1 12132-13132
00001:0 4 2 00001:0 12313-12323 ; 00001:0 4 3 00001:0 31212-31232
00011:0 0 1 00011:0 21230-23230 ; 00011:0 0 2 00112:0 12130-12132
00011:0 0 3 00112:0 12120-12130 ; 00011:0 1 1 00101:0 12103-12123
00011:0 1 2 00121:0 13103-23103 ; 00011:0 1 3 00121:0 12102-32102
00011:0 2 1 00011:0 12302-32302 ; 00011:0 2 2 00112:0 12103-12123
00011:0 2 3 00112:0 21230-31230 ; 00011:0 3 0 00001:0 12132-13132
00011:0 3 2 00012:0 12102-32102 ; 00011:0 3 3 00012:0 12102-32102
00011:0 4 0 00001:0 12123-12323 ; 00011:0 4 2 00012:0 12103-12123
00011:0 4 3 00012:0 12102-32102 ; 00012:0 0 1 00121:0 21203-23203
00012:0 0 2 00112:0 12320-32320 ; 00012:0 0 3 00123:0 12123-13123
00012:0 1 1 01012:0 12103-12123 ; 00012:0 1 2 01012:0 13103-13203
00012:0 1 3 01023:0 12103-12123 ; 00012:0 2 1 00112:0 12320-32320
00012:0 2 2 00121:0 12103-12303 ; 00012:0 2 3 00123:0 12103-12123
00012:0 3 0 00001:0 12130-13130 ; 00012:0 3 2 00011:0 12130-13130
00012:0 3 3 00012:0 12120-13120 ; 00012:0 4 0 00001:0 12123-13123
00012:0 4 1 00011:0 12103-12123 ; 00012:0 4 3 00012:0 12120-13120
00101:0 0 1 00101:0 21020-23020 ; 00101:0 0 2 01012:0 12010-32010
00101:0 0 3 01012:0 12013-12313 ; 00101:0 1 1 00101:0 12013-12313
00101:0 1 2 01012:0 13010-13020 ; 00101:0 1 3 01012:0 12012-32012
00101:0 2 0 00001:0 12310-32310 ; 00101:0 2 2 00102:0 12012-32012
00101:0 2 3 00102:0 12012-32012 ; 00101:0 3 1 00011:0 12023-13023
00101:0 3 2 00121:0 12010-32010 ; 00101:0 3 3 00121:0 12010-32010
00101:0 4 0 00001:0 12032-32032 ; 00101:0 4 2 00102:0 12010-32010
00101:0 4 3 00102:0 12010-32010 ; 00102:0 0 1 01012:0 21013-23013
00102:0 0 2 00121:0 12010-12030 ; 00102:0 0 3 01023:0 12310-12320
00102:0 1 1 00121:0 12013-12313 ; 00102:0 1 2 01012:0 13010-23010
00102:0 1 3 01023:0 12010-32010 ; 00102:0 2 0 00001:0 13213-23213
00102:0 2 2 00101:0 12010-32010 ; 00102:0 2 3 00102:0 12010-32010
00102:0 3 1 00112:0 12320-32320 ; 00102:0 3 2 00112:0 12310-32310
00102:0 3 3 00123:0 12010-32010 ; 00102:0 4 0 00001:0 21013-23013
00102:0 4 1 00101:0 12010-32010 ; 00102:0 4 3 00102:0 12010-12030
00112:0 0 1 00102:0 21030-21230 ; 00112:0 0 2 00112:0 12023-13023
00112:0 0 3 00123:0 12023-13023 ; 00112:0 1 1 00012:0 12023-13023
00112:0 1 2 00121:0 13020-13023 ; 00112:0 1 3 00123:0 12320-32320
00112:0 2 0 00012:0 12320-32320 ; 00112:0 2 2 00121:0 12023-13023
```

```
00112:0 2 3 00123:0 13020-13023 ; 00112:0 3 0 00102:0 12320-32320
00112:0 3 2 00112:0 13030-13230 ; 00112:0 3 3 00123:0 13020-13023
00112:0 4 0 00011:0 12023-13023 ; 00112:0 4 1 00011:0 12023-13023
00112:0 4 3 00112:0 12320-32320 ; 00121:0 0 1 00102:0 21013-23013
00121:0 0 2 01012:0 12010-32010 ; 00121:0 0 3 01023:0 12013-12313
00121:0 1 1 00102:0 12010-32010 ; 00121:0 1 2 01012:0 13030-13230
00121:0 1 3 01023:0 12013-12313 ; 00121:0 2 0 00012:0 12302-12303
00121:0 2 2 00112:0 12010-32010 ; 00121:0 2 3 00123:0 12010-32010
00121:0 3 0 00101:0 12010-32010 ; 00121:0 3 1 00011:0 12032-32032
00121:0 3 3 00121:0 12010-32010 ; 00121:0 4 0 00012:0 12012-13012
00121:0 4 2 00112:0 12013-13013 ; 00121:0 4 3 00123:0 12010-32010
00123:0 0 1 01023:0 21010-23010 ; 00123:0 0 2 01023:0 12010-12310
00123:0 0 3 00123:0 12010-12310 ; 00123:0 1 1 00123:0 12012-13012
00123:0 1 2 01023:0 13012-13032 ; 00123:0 1 3 01023:0 12030-12032
00123:0 2 0 00012:0 12302-32302 ; 00123:0 2 2 00112:0 12030-32030
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